

CAPSTONE PROJECT ON INSURANCE COST PREDICTION

DSBA

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List of Abbreviations

LR – Linear Regression
DTR – Decision Tree Regressor
RF – Random Forest
GB – Gradient Boost
XGB – Xtreme Gradient Boost
VIF – Variance Inflation Factor

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1 PROBLEM STATEMENT

The healthcare industry stands as a vital field where financial planning shapes individual well-being, because, at times, medical care can be extremely expensive, and if a person's insurance does not cover them, they may find themselves in a difficult financial situation. Medical insurance companies must accurately price premiums to match each applicant's health risk profile, walking the line between business stability and fairness to customers. This project takes on that challenge by developing a data-driven model that predicts insurance costs based on a person's health, demographic, and lifestyle attributes.

1.1 Objective

This project's goal is to create a model that uses data to determine an individual's optimal insurance cost. For the estimated cost of insurance, we must use the parameters associated with individuals habits and health.

1.2 Business & Social Opportunity

This project brings a real boost to both insurance companies and their customers. For insurers, the model means smarter risk assessment and clearer premium pricing. That way, they don't end up covering losses from high-risk clients. In contrast, customers get a premium that actually reflects their health and lifestyle choices. This will encourage the customers to adopt healthy habits such as regular exercise and health check-ups. From a larger perspective, this solution will help in the development of a more efficient healthcare insurance system that is accessible and affordable for the customers.

1.3 Data Description

The data covers 24 features associated with the customer availing insurance. Below is a detailed data dictionary.

Feature	Description
applicant_id	Applicant unique ID (Unique Identifier)
years_of_insurance_with_us	Since how many years customer is taking policy from the same company only
regular_checkup_lasy_year	Number of times customers has done the regular health check-up in last one year.
adventure_sports	Customer is involved with adventure sports like climbing, diving etc.(Binary)
Occupation	Occupation of the customer (category: Salried, Student, or Business)
visited_doctor_last_1_year	Number of times customer has visited doctor in last one year
cholesterol_level	Cholesterol level of the customers while applying for insurance
daily_avg_steps	Average daily steps walked by customers age of the customer
Age	Age of the customer
heart_decs_history	Any past history of heart disease (Binary)
other_major_decs_history	Any past major diseases other than heart like any operations (Binary)
Gender	Gender of the customer
avg_glucose_level	Average glucose level of the customer while applying the insurance

Feature	Description
bmi	BMI of the customer while applying the insurance
smoking_status	Smoking status of the customer (category: never smoked, formerly smoked, smokes, unknown)
Year_last_admitted	When customer have been admitted in the hospital last time
weight	Weight of the customer
covered_by_any_other_company	Whether customer is covered by another insurer (Binary)
Alcohol	Alcohol consumption status of the customer (category: No, Rare, Daily)
exercise	Regular exercise status of the customer (category: No, Moderate, Extreme)
weight_change_in_last_one_year	Weight change of the customer in the last one year
fat_percentage	Fat percentage of the customer while applying the insurance
insurance_cost	TARGET: Total insurance cost

Table 1. Data Description

1.4 Data Collection & Background

The data set utilized in this project comprises **25,000 data points with 24 features**, focusing on various aspects of an individual's demographic information, medical history, and lifestyle. The dataset source is not disclosed; it assumed to be from a system that provides insurance to its customers, and it is useful in determining various factors that affect insurance costs. The dataset has both **numerical (16) and categorical (8) variables**, which is used for modelling.

2 DATA OVERVIEW

Methodology: Initial inspection was performed on dataset to understand the its structure, variable types, summary statistics, and data quality issues before proceeding for detailed analysis. Dataset's detailed inspection can be viewed in the Appendix section.

Displaying top 5 rows of dataset

	applicant_id	years_of_insurance_with_us	regular_checkup_last_year	adventure_sports	Occupation	visited_doctor_last_1_year	cholesterol_level	daily_avg_s
0	5000	3	1	1	Salaried	2	125 to 150	
1	5001	0	0	0	Student	4	150 to 175	
2	5002	1	0	0	Business	4	200 to 225	
3	5003	7	4	0	Business	2	175 to 200	
4	5004	3	1	0	Student	2	150 to 175	

5 rows × 24 columns

Table 2. Top 5 rows

Statistical Summary

	count	mean	std	min	25%	50%	75%	max
applicant_id	25000.0	17499.500000	7217.022701	5000.0	11249.75	17499.5	23749.25	29999.0
years_of_insurance_with_us	25000.0	4.089040	2.606612	0.0	2.00	4.0	6.00	8.0
regular_checkup_last_year	25000.0	0.773680	1.199449	0.0	0.00	0.0	1.00	5.0
adventure_sports	25000.0	0.081720	0.273943	0.0	0.00	0.0	0.00	1.0
visited_doctor_last_1_year	25000.0	3.104200	1.141663	0.0	2.00	3.0	4.00	12.0
daily_avg_steps	25000.0	5215.889320	1053.179748	2034.0	4543.00	5089.0	5730.00	11255.0
age	25000.0	44.918320	16.107482	16.0	31.00	45.0	59.00	74.0
heart_decs_history	25000.0	0.054640	0.227281	0.0	0.00	0.0	0.00	1.0
other_major_decs_history	25000.0	0.008160	0.297537	0.0	0.00	0.0	0.00	1.0
avg_glucose_level	25000.0	167.530000	62.729712	57.0	113.00	168.0	222.00	277.0
bmi	24010.0	31.393328	7.876535	12.3	26.10	30.5	35.60	100.6
Year_last_admitted	13119.0	2003.892217	7.581521	1990.0	1997.00	2004.0	2010.00	2018.0
weight	25000.0	71.610480	9.325183	52.0	64.00	72.0	78.00	96.0
weight_change_in_last_one_year	25000.0	2.517960	1.690335	0.0	1.00	3.0	4.00	6.0
fat_percentage	25000.0	28.812280	8.632382	11.0	21.00	31.0	36.00	42.0
insurance_cost	25000.0	27147.407680	14323.691832	2468.0	16042.00	27148.0	37020.00	67870.0

Table 3. Statistical Summary

Key Observations from Data Overview

- Our dataset has **25,000 rows and 24 columns**.
- There are **8 categorical variables and 16 numerical variables**.
- No duplicate entries detected.
- **Few columns have missing values and spelling mistakes.**
- Features include demographic, clinical, and behavioral variables.
- We dropped the applicant_id because it is a unique identifier.

After deleting the identifier column, 23 features remained. The initial review also revealed missing values in BMI and Year_last_admitted, a large “Unknown” smoking category, and other data quality issues requiring pre-processing before modelling.

3 EXPLORATORY DATA ANALYSIS

Methodology: EDA was carried out using univariate, bivariate, and multivariate analysis to comprehend variable distributions, identify relevant patterns, and analyse correlations with insurance costs. Only the most significant visuals are shown here; full EDA plots are in the Appendix.

3.1 EDA Summary Table

Feature	Mean	Std. Dev	Key Observation
insurance_cost (target)	27,147.41	14,323.69	Multimodal data; no outliers; Value range: 2,468–67,870
weight	71.61	9.33	It is the strongest predictor; balanced; no outliers
bmi	31.39	7.88	Distribution is right-skewed; max ~100; outliers present
age	44.92	16.00	Distribution is uniform, age range: 16 to 74; near-zero correlation with cost.

Feature	Mean	Std. Dev	Key Observation
avg_glucose_level	167.50	62.73	Wide range 57–277; distribution is symmetric; no outliers present
daily_avg_steps	5,216	1,053	Moderate activity; 952 high-step outliers retained
visited_doctor_last_1_year	3.10	1.14	right-skewed distribution; outliers present
regular_checkup_lasy_year	0.77	1.13	Heavily right-skewed distribution; 60.9% had zero check-ups
fat_percentage	28.81	8.63	Mildly left skewed distribution; outliers present: are realistic data (11-42%)
weight_change_in_last_one_year	2.52	1.69	Multi-modal distribution ; no outliers present
Years_of_insurance_with_us	4.23	2.62	Uniform distribution with range 0–8 years;

Table 4. Key EDA Summary :Numerical features

3.2 Univariate Analysis

Univariate analysis was performed on all numerical and all categorical columns to understand the their distributions, trends and identify anomalies like outliers.

Distribution of insurance_cost (Target variable)

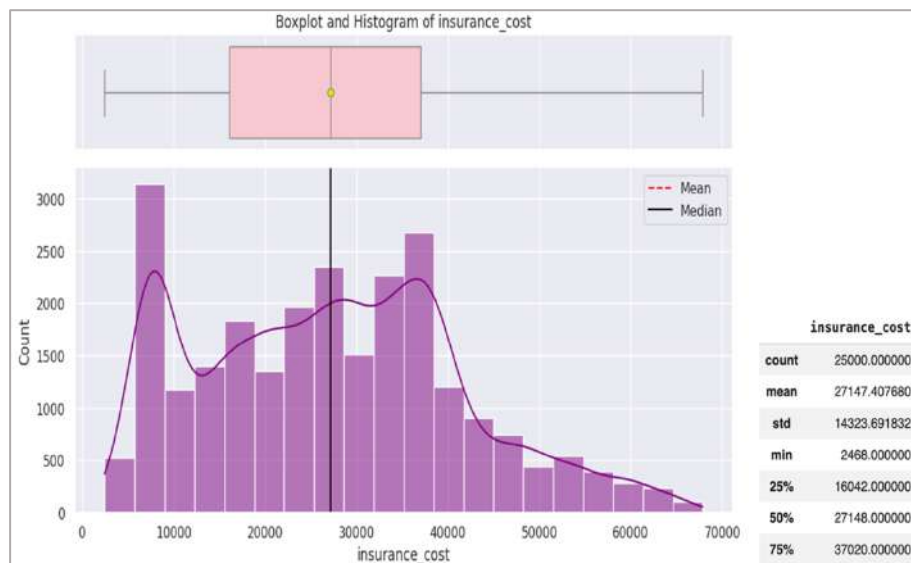


Figure 1. Distribution of insurance cost

Observations:

- Average insurance cost $\approx 27,147.41$, with a std dev $\approx 14,323.69$, Range $\approx 2,468 - 67,870$.
- The plot shows a multimodal distribution with multiple peaks, suggests distinct customer risk profile.
- The boxplot also shows no outliers.

Distribution of Weight (Primary Predictor)

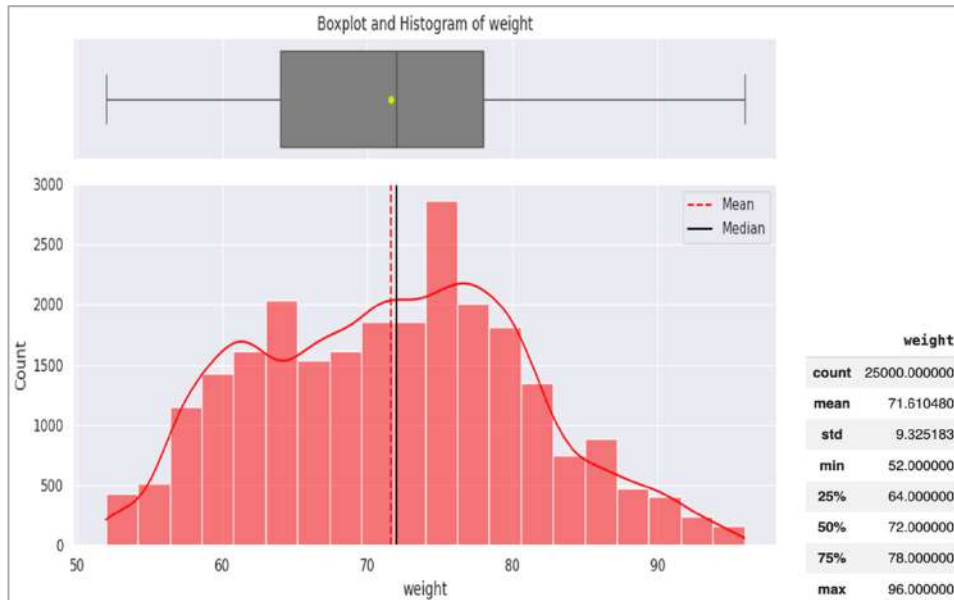


Figure 2. Distribution of weight

Observations:

- Average weight ≈ 71.61 , with a std dev ≈ 9.33 , Weight range $\approx 52 - 96$.
- There is a slight skewness to the left; however, the majority of the individuals fall between 64 units and 78 units (interquartile range).
- There are no statistical outliers . Multiple peaks suggests demographic sub-populations.

regular_checkup_last_year

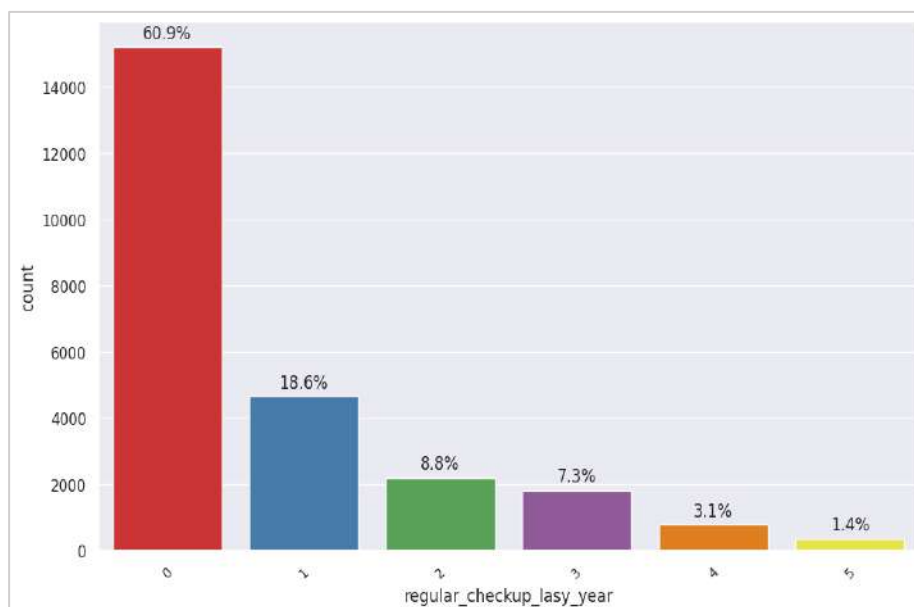


Figure 3. Distribution of regular_checkup_last_year

Observations:

- Average weight ≈ 0.77 , with a std dev ≈ 1.13 , check-ups range $\approx 0-5$.
- There is significant skewness to the right, indicating that most individuals have fewer than 5 check-ups, with the vast majority of individuals (about 79.5% of the total population) having either 0 or 1 check-ups.

Distribution of covered_by_any_other_company

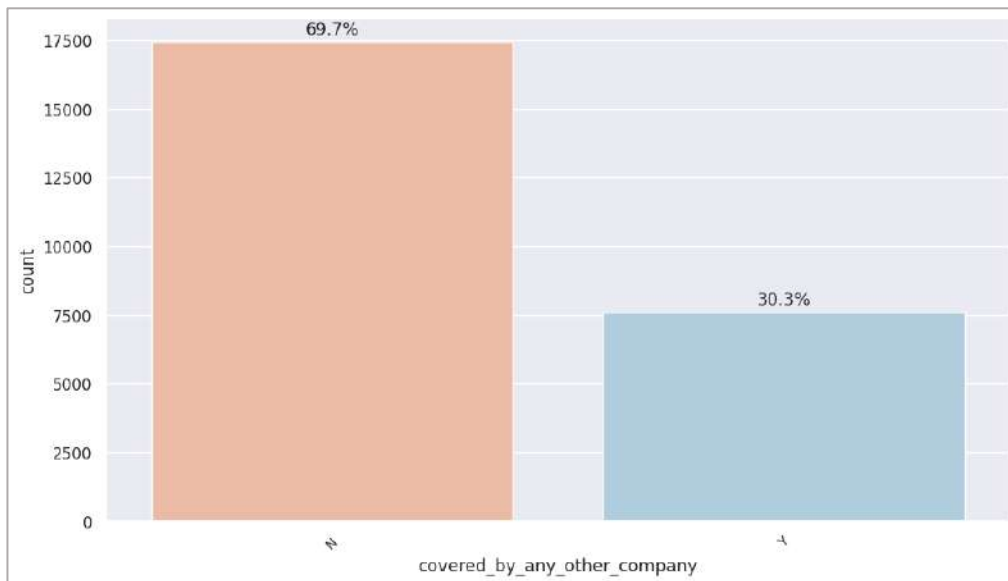


Figure 4. Distribution of covered_by_any_other_company

Observations:

- It is a binary variable(Y/N), i.e., whether or not the applicants have overlapping coverage with a secondary insurance provider.
- Approximately 70% of the population does not have external coverage, while approximately 30% of the population holds other companies' insurance policies.
- Class distribution is highly skewed, as the number of individuals without coverage being over twice as large as the number of individuals with coverage.

3.3 Bivariate Analysis

The bivariate analysis explored relationship between individual features and target variable (insurance_cost).

Distribution of weight vs Insurance Cost (Target vs Numerical Column)

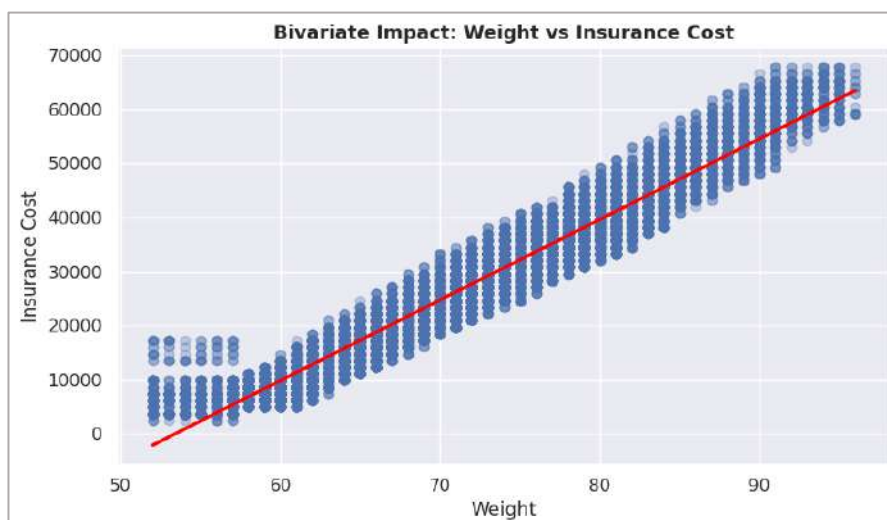


Figure 5. Regression plot: weight vs insurance_cost

Observations:

- The presence of a strong positive linear correlation is evident between weight and insurance cost, where cost increases linearly with increasing weight from 52 to 96.
- Linear increasing weight, reflects a stable insurance cost structure.

Smoking Status vs Insurance Cost (Categorical vs Target Column)

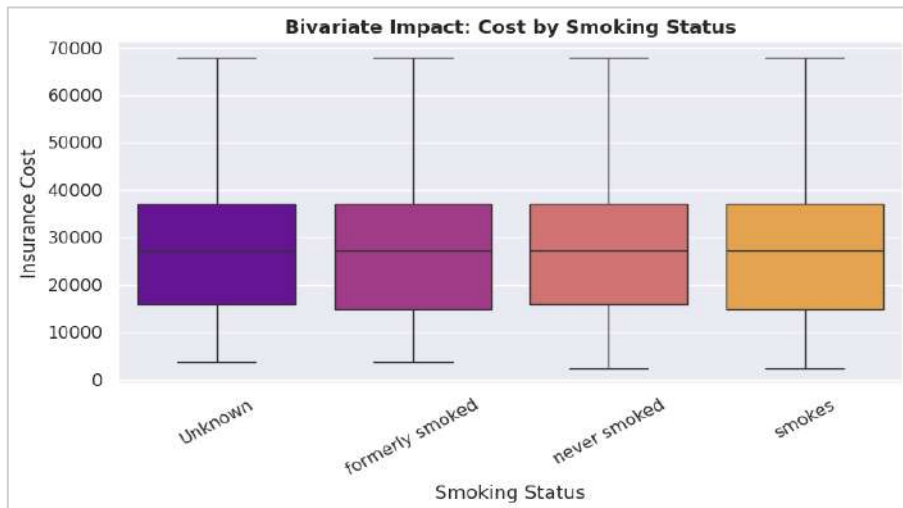


Figure 6. Distribution of smoking_status vs insurance_cost

Observations:

- Insurance costs for all smoking categories range roughly from 3000 to 68000.
- No meaningful cost difference across all smoking categories (smokers, never smokers, former smokers, and unknown smokers)
- There is no discernible trend across all groups; smoking status is not a strong predictor.

Occupation vs Exercise (Categorical vs Categorical Column)

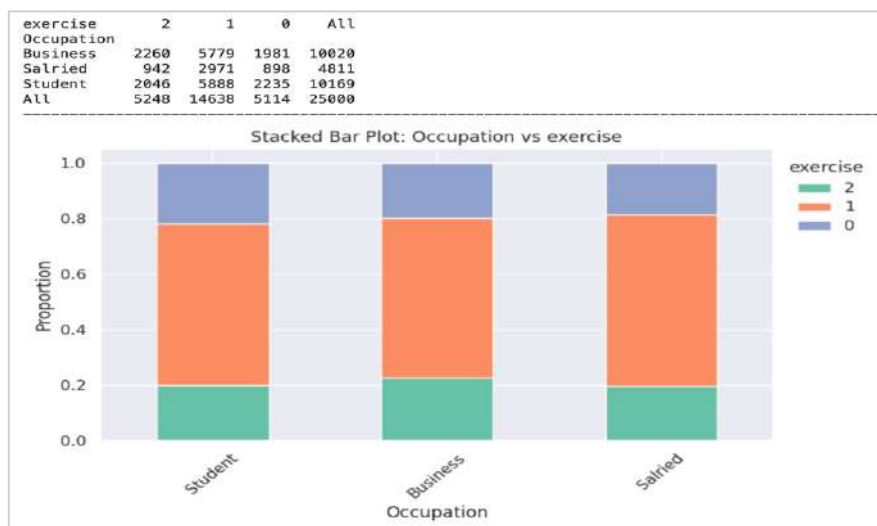


Figure 7. Stacked bar plot of Occupation vs exercise

Observations:

- Exercise patterns are nearly uniform across different occupations.
- The majority of individuals in all groups participate in moderate exercise (level 1), which is a similar trend across the board.
- It appears that occupation is not a significant factor in determining the exercise pattern.

3.4 Multivariate Analysis

Correlation Analysis using Heatmap

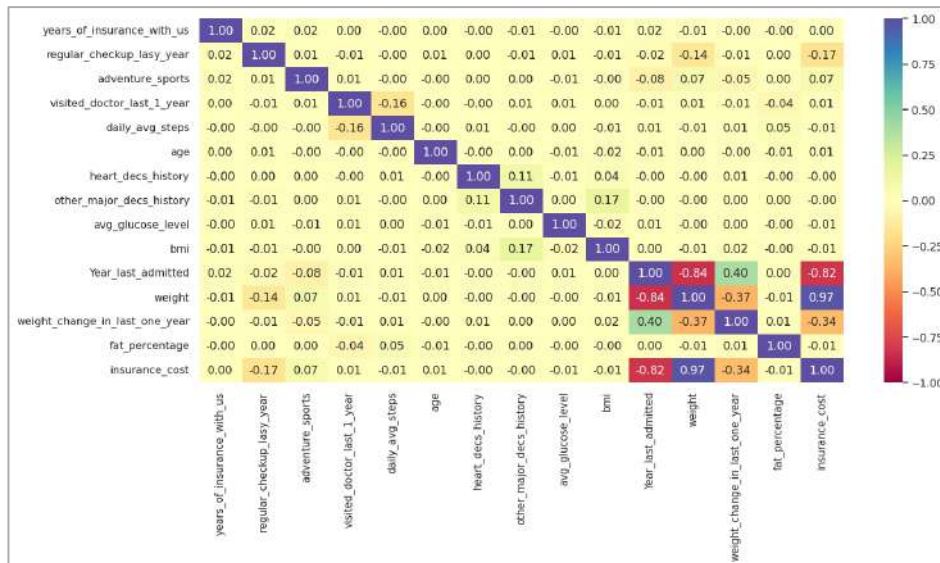


Figure 8. Correlation Heatmap of all Numerical variables

Observations:

- Weight has the strongest positive relationship with insurance cost, whereas the majority of the other factors have a near-zero linear correlation.
- Year_last_admitted has a strong negative correlation, supporting feature engineering decision.

Density Plot of Average Glucose Level vs Insurance Cost vs Heart Disease (Interaction Effect)

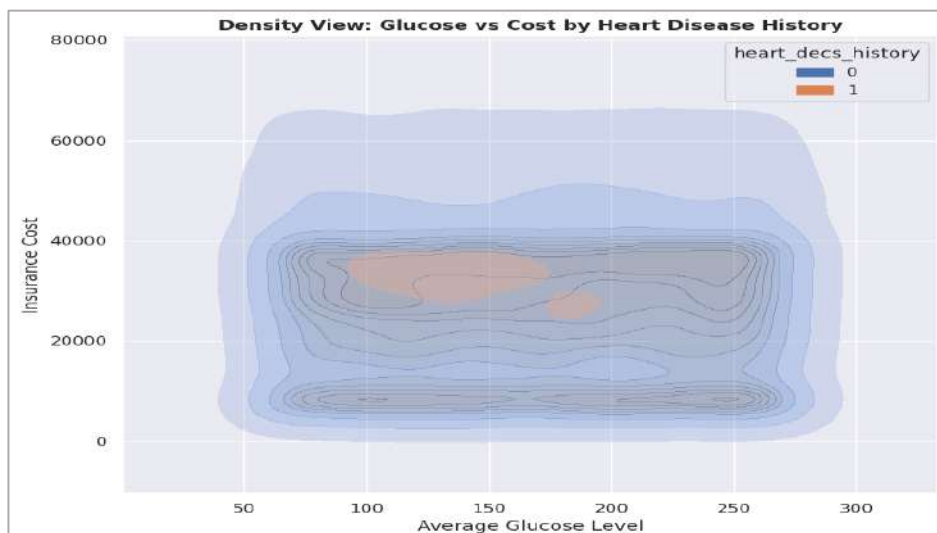


Figure 9. Density Plot: Avg_glucose_level vs insurance_cost vs heart_decs_history

Observations:

- Individuals with heart disease show tighter and higher-cost clusters, while customers without heart disease have a wide cost and glucose range.
- This shows that heart disease history may be important in combination with other factors, even if average glucose level alone is not a strong predictor.

Scatter Plot Of Weight vs. Insurance Cost by Smoking Status (Interaction Effect)

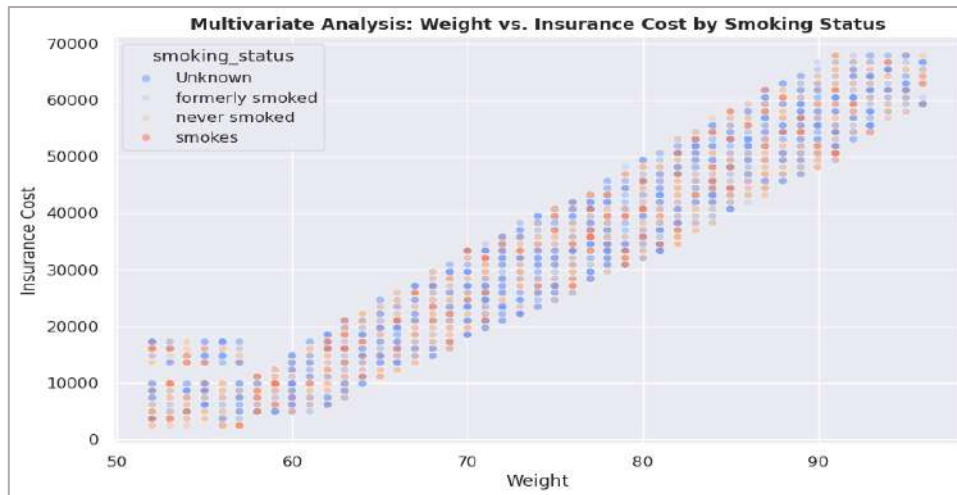


Figure 10. Scatter plot: weight vs insurance_cost by smoking_status

Observations:

- The relationship between weight and insurance costs is highly positive across all smoking groups.
- The smoking groups substantially follow the same trend line, demonstrating that weight is the most significant factor even when smoking status is accounted for.

3.5 Overall Insights from EDA

- **Weight** appeared as the **most powerful predictor** of insurance costs, with a very strong positive correlation to the target variable.
- Most other **health, lifestyle, and geographic variables** had poor standalone correlations with insurance costs, indicating **minimal independent predictive potential**.
- The dataset also revealed **significant structural flaws**, such as missing values in BMI and Year_last_admitted, a big "Unknown" category in smoking status, and an imbalance in some variables.
- Some observed trends, such as the near-zero correlation between age and insurance cost, appear to contradict real-world underwriting logic, implying that the dataset may not accurately reflect true pricing behaviour.
- These findings emphasised the importance of **missing value treatment, selective outlier handling, and feature engineering** in improving the dataset's usability for modelling.

4 DATA PRE-PROCESSING

Methodology: Data pre-processing was performed to clean, transform, engineer features to prepare the dataset for modelling. Refer the appendix for detailed pre-processing outputs.

4.1 Data Cleaning

- **No duplicate** values detected.
- **Corrected naming inconsistencies:** 'Salried' to 'Salaried' in the occupation column.
- **Fixed spelling error:** 'regular_checkup_lasy_year' to 'regular_checkup_last_year'.
- **Dropped 'Location'** column, prior EDA showed no meaningful pricing variation across cities; retaining it lead to noise and keep the algorithm from learning misleading patterns.
- **After cleaning**, the shape of our dataset was 25000 rows and **22 columns**.

4.2 Anomalous Value

- **No anomalous records were found** in any of the domain-critical checks; the dataset does not contain any negative ages, weights, glucose levels, or impossible fat percentages.
- **89 records with BMI > 60** were flagged, these are clinically valid (Super Obesity case) and these were retained. Therefore, these records were addressed during the outlier treatment phase using Winsorization to avoid selection bias.

4.3 Missing Values & Treatment

Missing values:	
	0
years_of_insurance_with_us	0
regular_checkup_lasy_year	0
adventure_sports	0
Occupation	0
visited_doctor_last_1_year	0
cholesterol_level	0
daily_avg_steps	0
age	0
heart_decs_history	0
other_major_decs_history	0
Gender	0
avg_glucose_level	0
bmi	990
smoking_status	0
Year_last_admitted	11881
Location	0
weight	0
covered_by_any_other_company	0
Alcohol	0
exercise	0
weight_change_in_last_one_year	0
fat_percentage	0
insurance_cost	0

Table 5. Missing values counts by column

- **bmi: 990 missing** values (~ 4%), treated via median imputation grouped by gender.
- **Year_last_admitted: 11,881 missing** values (~ 47.5%); **feature engineered** rather than imputed:
 - **ever_hospitalized** (binary 0/1): Indicates if patient was **ever hospitalized** (1 if year exists, 0 if NaN). It is of int data type.

- **years_since_last_admission** (numeric): **Years elapsed** since last admission using 2018 as reference (2018 - Year_last_admitted, 0 if never hospitalized). It is of int data type.
- **Dropped "Year_last_admitted"** to avoid redundancy.
- Post -treatment dataset was found to have zero remaining missing values.

4.4 Outlier Detection & Treatment

Methodology: Outlier detection applied only to continuous numerical features via IQR & Box plot methods for ensuring the data quality.

Outlier Check

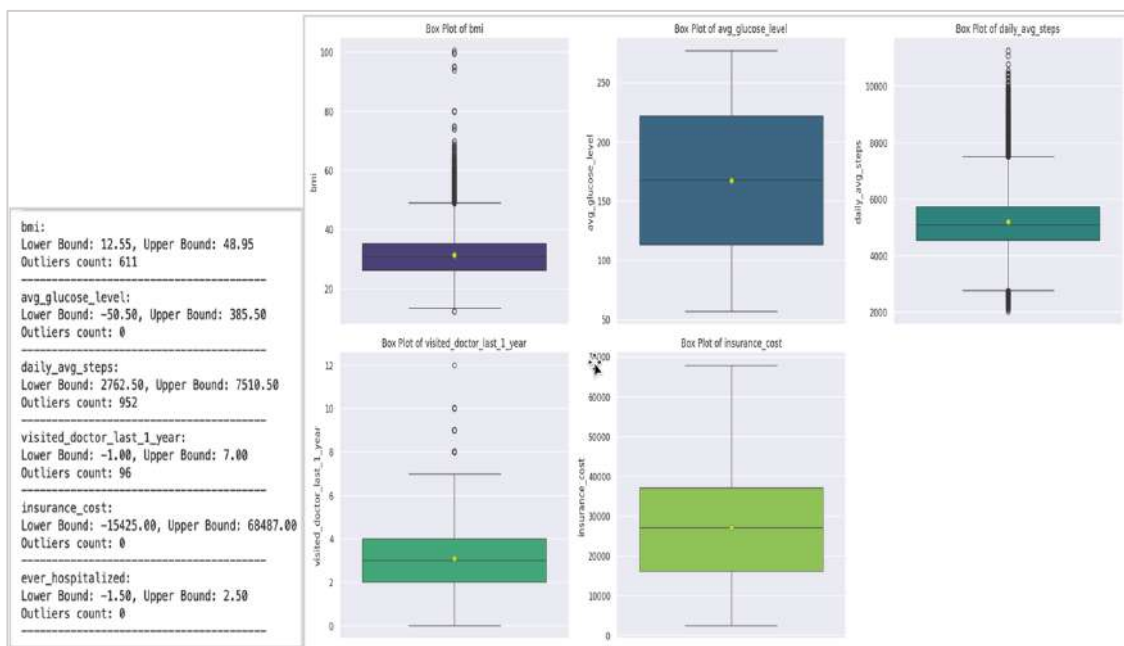


Figure 11. Outlier detection-boxplot for all numerical features

Feature	Outliers Detected	Treatment Decision	Rationale
bmi	611 outliers (right skewed, max ~100)	IQR clipping: capped at 12.55 (min)–48.95 (max)	Clinically valid but extreme; capping reduces impact of extreme data points while preserving distribution
daily_avg_steps	952 outliers	No treatment required	represent genuine behavioural variation; realistic pattern (high activity/sedentary)
visited_doctor_last_1_year	96 outliers (>7 visits)	No treatment required	genuine high-frequency patients with complex medical conditions
avg_glucose_level	No outliers	No treatment required	All values falls within IQR-based bounds naturally
insurance_cost	No outliers	No treatment required	target variable; values would altering it would change the prediction objective

Table 6. Outlier Treatment Decisions

Post-treatment verification revealed **no remaining outliers**, keeping the distribution within statistically acceptable bounds.

4.5 Feature Engineering

Methodology: Raw features were transformed into more informative, enabling the model to capture the underlying clinical signals and improve predictive accuracy & power.

- **Validated ‘ever_hospitalized’ feature:** Before using ‘ever_hospitalized’ column, we ensured it represents a relevant medical subgroup rather than random missingness.
- **Created “health_risk_score”:** We combined “heart_decs_history”, “other_major_decs_history”, and “ever_hospitalized” into one feature into medical risk score (0-3) representing entire health risk profile of customers for providing better cost prediction.

EDA on Engineered Features

We did further analysis on engineered features (ever_hospitalized, years_since_last_admission, health_risk_score) to validate whether they show a meaningful variation in insurance cost (target variable) before model training.

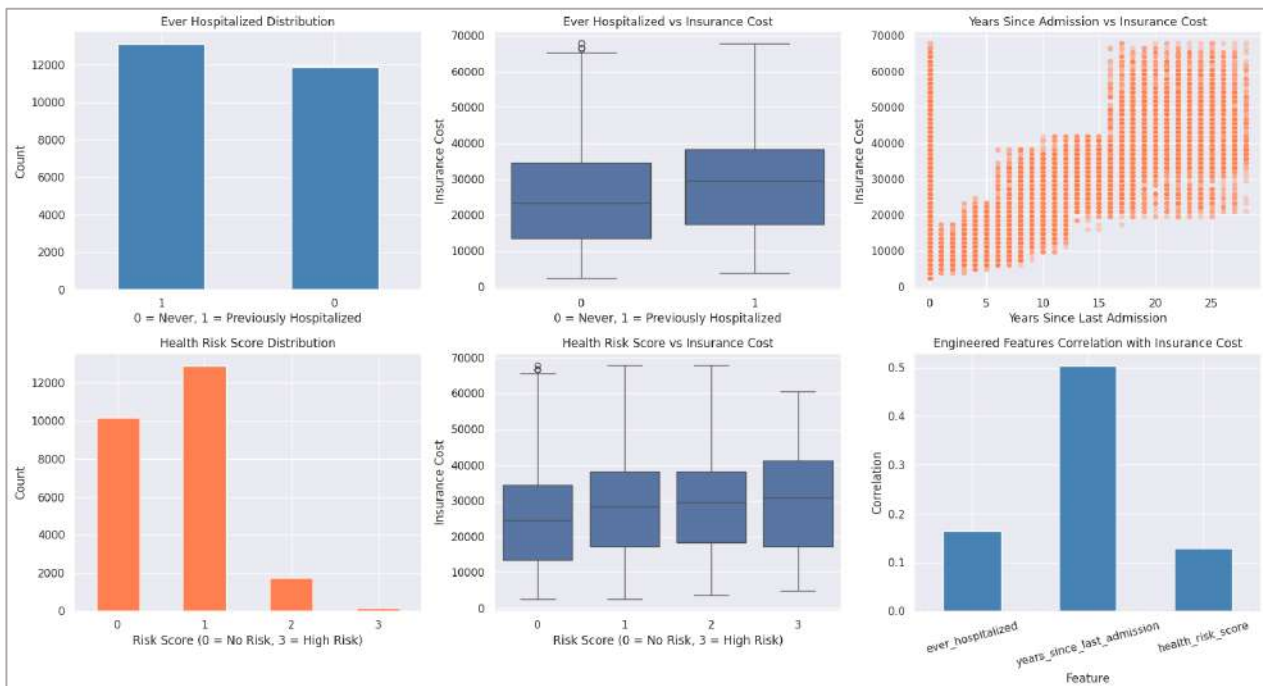


Figure 12. EDA on Engineered Features

- **Years_since_last_admission:** is the most influential engineered feature, with the strongest correlation (~0.50) and a noticeable upward trend in the scatter plot.
- **ever_hospitalized:** previously hospitalised clients show higher and more variable insurance costs.
- **health_risk_score:** steady rise in insurance costs as risk level grows (0-3), suggesting that integrating numerous health indicators into a single feature yields a valuable aggregated signal.

- Overall, the engineered features collectively capture temporal (years), binary (hospitalization status), and cumulative (risk score) aspects of health risk, improving the dataset's potential to represent patterns which are relevant for modeling.

4.6 Class Imbalance Handling

Since the target variable is continuous & variable in nature (a regression problem), indicating **class imbalance handling is not applicable, and no treatment required in this case**, because class imbalance handling is applied exclusively to classification problems where target variables are discrete and potentially unequal in frequency.

4.7 Data Preparation for Modeling

Methodology: In order to prepare the data so that machine learning algorithms could accurately and without bias process the raw clinical data, we used following steps: **encoding, train/validation/test, feature scaling** and **data leakage prevention**.

- **Encoding:** Smoking status category 'unknown' (over 30% of data) was **first replaced with NaN** value and then **imputed with mode** values rather than dropping to preserve the data integrity. We **mapped 'never smoked': 0, 'formerly smoked': 1, and 'smokes': 2**. **One-hot encoding** was applied to 'Occupation' column. After encoding, final dataset dimensions were 25,000 rows and 25 columns.
- **Data Splitting :** Initial baseline model used an **80:20** train-test split. The final baseline and advanced models used **60:20:20 split** (15,000 train / 5,000 validation / 5,000 test) for consistent cross-model performance comparison.
- **Feature Scaling:** StandardScaler was applied to training data only; then used to transform both validation and test sets ((mean $\approx$ 0, std $\approx$ 1). Initial baseline OLS model used unscaled data, while final baseline and advanced model used scaled data to preserve coefficient interpretability in business units.
- **Data Leakage Handling:** Scaler fitted only on training & validation data then later on test set during final predictions. Engineered features derived from domain knowledge & EDA. All pipeline integrity checks passed.
- Since, dataset was thoroughly pre-processed, encoded, split, scaled, and leakage-free, making it ideal for baseline model building.

5 MODELING APPROACH & MODEL SELECTION

Since target variable (insurance_cost) is a continuous numerical variable, this is a regression problem. A **linear regression** baseline was selected its simplicity, interpretability. However, EDA revealed non-linear patterns in the data, motivating the use of advanced tree-based models such as **Decision Tree, Random Forest, Gradient Boosting, XGBoost** and **AdaBoost**. Ensemble methods were selected to capture complex relationships and improve prediction accuracy over the baseline.

5.1 Metric of Choice

The **primary evaluation metric RMSE (Root Mean Squared Error)** is selected because:

- Insurance pricing is risk-sensitive and large prediction errors directly cause financial loss.
- RMSE penalizes large errors more heavily than MAE
- Same unit as insurance_cost and it directly interpretable in business terms
- Aligns with the core objective of minimizing costly mispricing's.

The supporting metrics were used for more comprehensive evaluation. **MAE** (average error magnitude) as it is less sensitive to outliers, helping to understand typical prediction deviations. **R-squared-Adjusted R²** (variance explained), indicates the proportion variance explained by the model, while **Adjusted R²** accounts for number of predictors, offering insights into overall model fit after considering feature complexity. **MAPE** expresses error in percentage terms, making it easy for business interpretation, although it is used cautiously due to the presence of relatively low cost values in dataset.

6 BASELINE MODEL BUILDING – LINEAR REGRESSION

Full OLS (Ordinary Least Squares) statistical summaries, coefficient tables, multicollinearity check and assumption check outputs of all the models are present in the Appendix.

6.1 Initial Baseline (OLS) Model

Methodology

To establish an initial benchmark for predicting insurance cost accurately, (OLS) Linear Regression approach was implemented on training data. Unscaled data was used in this model for coefficient interpretability in original business units, enabling meaningful insights for stakeholders. Key statistical measures (R², Adjusted R², p-values, F-statistic) were assessed to analyse predictive performance and statistical reliability. Metrics like RMSE, MAE, and MAPE were used to evaluate both training performance and generalisation on test data (only on initial model).

Key Observation from OLS Summary

- The model accounts for approximately **94.5% of the variance in insurance costs**, suggesting good predictive performance.
- The F-statistic indicates that the model as a whole is **statistically significant**. Several predictors are statistically significant, while others contribute only marginally to the model.
- **Weight emerges as the most relevant factor**, with a strong positive correlation to insurance costs.
- The model has a **high condition number**, indicating **strong multicollinearity** among variables.
- Since baseline model has high prediction accuracy but suffers from multicollinearity and feature redundancy. Therefore, more refining techniques such as VIF-based feature selection is required to improve model stability and interpretation.

Initial Baseline Model Coefficients

	Feature	Coefficient
0	const	-78199.318358
1	years_of_insurance_with_us	-89.805426
2	regular_checkup_last_year	-375.711524
3	adventure_sports	227.832251
4	visited_doctor_last_1_year	-51.563685
5	cholesterol_level	23.117579
6	daily_avg_steps	-0.015862
7	age	4.135229
8	heart_decs_history	198.328841
9	other_major_decs_history	-61.737727
10	Gender	-23.130393
11	avg_glucose_level	0.135723
12	bmi	-3.111962
13	smoking_status	-27.346372
14	weight	1465.439136
15	covered_by_any_other_company	1197.691991
16	Alcohol	-3.978872
17	exercise	-21.148345
18	weight_change_in_last_one_year	173.125909
19	fat_percentage	-1.948896
20	ever_hospitalized	-54.490674
21	years_since_last_admission	51.260414
22	health_risk_score	82.100441
23	Occupation_Salaried	87.753090
24	Occupation_Student	45.114167

Table 7. Initial Baseline Model Coefficient table

- **Weight** has the **highest positive coefficient**, suggesting weight as **dominant driver of costs**.
- Individuals **covered by any other company** shows a significant positive coefficient, indicating that those who are already covered by another insurance may be considered as higher risk.
- **Regular health check-ups** shows a negative coefficient, indicating better health management associates with lower insurance costs.
- **Longer insurance tenure** are associated with slightly lower costs, potentially due to loyalty or lower perceived risk or customer loyalty.
- The coefficient analysis shows that both **lifestyle and health-related features** have a **considerable influence on insurance costs**. However, it seems because of the presence of multicollinearity, these coefficients may not be totally reliable, necessitating to do refinement with VIF.

Initial Baseline Model Performance Evaluation

Original Model Performance:					
	RMSE	MAE	R-squared	Adj. R-squared	MAPE (%)
Train	3350.446476	2697.218392	0.945379	0.945313	15.057427
Test	3352.451619	2707.200266	0.944835	0.944569	15.297431

Table 8. Initial Baseline Model Performance Evaluation

- $R^2 \approx 0.945$, the model explains $\sim 94.5\%$ of the variance in insurance costs. suggesting strong predictive power.
- **RMSE $\approx 3,350$** on both the train and test sets indicates consistent performance, no overfitting.
- **The MAE (~ 2700); MAPE ($\sim 15\%$)** reflects moderate percentage error across the observations.
- The initial linear regression model has **strong predictive performance**, high predictive power, but have **multicollinearity** issue, model refinement is required to increase model stability and interpretability.

6.2 Refined Baseline Model-Linear Regression

After building initial baseline model we need to refine it further in order to improve our model's performance.

6.2.1 Multicollinearity Check Using VIF Scores

Methodology: VIF analysis performed on training dataset only, to ensure proper model validation. The constant (intercept) term was excluded from the calculation, as it is not an actual predictor. Variables with high VIF > 5 were flagged. Perfect multicollinearity resolved by dropping redundant features. Here VIF is used as feature selection tool. This phase comes before p-value analysis because multicollinearity can have an impact on statistical significance.

	Feature	VIF
7	heart_decs_history	inf
21	health_risk_score	inf
19	ever_hospitalized	inf
8	other_major_decs_history	inf
13	weight	52.510703
5	daily_avg_steps	23.343937
11	bmi	20.980769
18	fat_percentage	12.712078
6	age	8.485722
3	visited_doctor_last_1_year	8.206053
10	avg_glucose_level	7.834353
20	years_since_last_admission	6.260945
0	years_of_insurance_with_us	5.431668
17	weight_change_in_last_one_year	3.482006
16	exercise	3.467633
4	cholesterol_level	2.848257
23	Occupation_Student	2.727572
15	Alcohol	2.526954
9	Gender	1.781805
1	regular_checkup_lasy_year	1.622965
22	Occupation_Salaried	1.607326
14	covered_by_any_other_company	1.564566
12	smoking_status	1.482135
2	adventure_sports	1.098058

Table 9. VIF score- initial baseline model

- VIF was used to identify **perfect multicollinearity** (infinite values) caused by including both raw medical history data and the engineered health_risk_score. In order to handle multicollinearity, we **systematically dropped redundant columns** with high multicollinearity and low statistical significance.
- After removing the redundant raw variables, perfect collinearity was achieved. The **condition number dropped** from 7.28×10^{18} to approximately 950, suggesting a **stable design matrix**.
- For the remaining variables with very high VIF > 10 (e.g., "weight", "age", etc.), suggests **natural health correlations** and does bias predictions.

6.2.2 Building Refined Model & Performance Evaluation.

Methodology: After dropping the high VIF & high p-value (using backward elimination, p-value > 0.05), i.e., statistically insignificant features were eliminated. Initial baseline model was re-fitted to obtain refined model and performance was evaluated on train set only.

Refined Model Performance (Train Data Only):					
	RMSE	MAE	R-squared	Adj. R-squared	MAPE (%)
0	3351.531679	2698.380311	0.945344	0.945319	15.063243

Table 10. Refined Model Performance Evaluation

- A total of **9 features were eliminated**, including lifestyle, demographic, and weak clinical variables.
- The **final 9 significant features**: 'years_of_insurance_with_us', 'regular_checkup_last_year', 'adventure_sports', 'visited_doctor_last_1_year', 'age', 'weight', 'covered_by_any_other_company', 'weight_change_in_last_one_year', 'years_since_last_admission', all with p-values less than 0.05.
- The Adjusted R-squared was slightly increased than original model. This shows that the eliminated variables made no substantial contribution to the model.

- $R^2 = 0.9453$, virtually unchanged from initial baseline (0.9450) , demonstrating that the model maintains its strong predictive power even after features removal.
- The **RMSE rose marginally** by only 1.08 (0.03%), statistically insignificant.
- Similarly, MAE increased somewhat from 2697.22 to 2698.38, showing a little shift in average prediction variance.
- The **MAPE** stayed nearly constant ($\approx 15.06\%$), suggesting that the model's percentage error performance is stable.
- The relatively minor changes in all performance metrics show that the eliminated features made no significant contribution to prediction performance.
- Therefore, **refined model retains only statistically significant predictors**, making model simpler, more interpretable with maintained predictive accuracy.

6.2.3 Refined Model - Assumptions Check

Methodology: 5 key linear regression assumptions were tested: No multicollinearity, linearity of variables & independence of error terms, normality of error terms, no heteroscedasticity. Detailed outputs are in the Appendix. Detailed methodology, plots and outputs for each assumption are present in the Appendix

Test for Multicollinearity

final VIF values for refined model:		
	Feature	VIF
5	weight	16.889288
4	age	8.254900
3	visited_doctor_last_1_year	7.983753
0	years_of_insurance_with_us	4.408905
7	weight_change_in_last_one_year	3.070130
8	years_since_last_admission	2.608119
6	covered_by_any_other_company	1.556804
1	regular_checkup_lasy_year	1.510419
2	adventure_sports	1.096862

Table 11. Multicollinearity Assumption Check

- After dropping high p-values, "weight" has now a VIF of 16.89, suggests **natural correlation** with other health and lifestyle features (e.g., age, doctor visits, etc) and does not bias predictions.
- All other predictors have low to moderate multicollinearity range (e.g., age, visited_doctor_last_1_year, etc).
- The **assumption of no perfect multicollinearity is satisfied** as the design matrix is full-rank and the condition number is 956.

Test for Linearity of Variables & Independence of Residuals

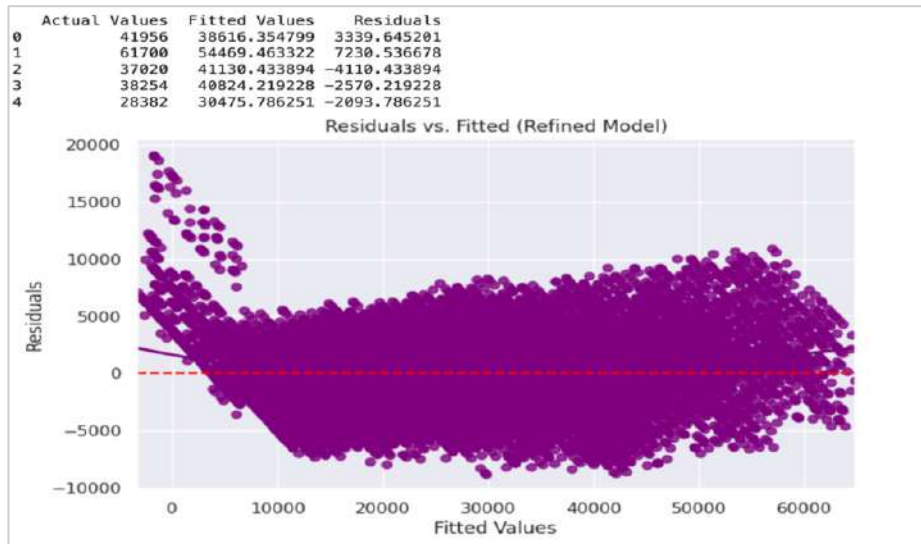


Figure 13. Plot of Residual vs Fitted values - refined baseline model

- The residuals are centred around zero and have no systematic curvature, which suggests no non-linearity.
- **Horizontal red line ($Y=0$)** shows No bias
- A few larger residuals appear at the extremes, but they do not dominate the distribution.
- The plot indicates that the refined model accurately depicts the linear relationship between variables and insurance costs.
- We see **no pattern** in the plot above. Hence, the **assumptions of linearity and independence are satisfied**.
- The Durbin-Watson statistic ≈ 1.97 (refined model OLS summary), indicates **no significant autocorrelation**.

Test for Normality of Residuals

- **Normality of Residuals Plot:** The **distribution of residuals was roughly symmetric and bell-shaped**, with the bulk of observations near zero. but a slight positive skew is present and the right tail extends to $\sim 19,000$, while the left tail is short ($\sim -5,000$) and contains very few points. This matches the skewness value of 0.367 reported in the model summary.

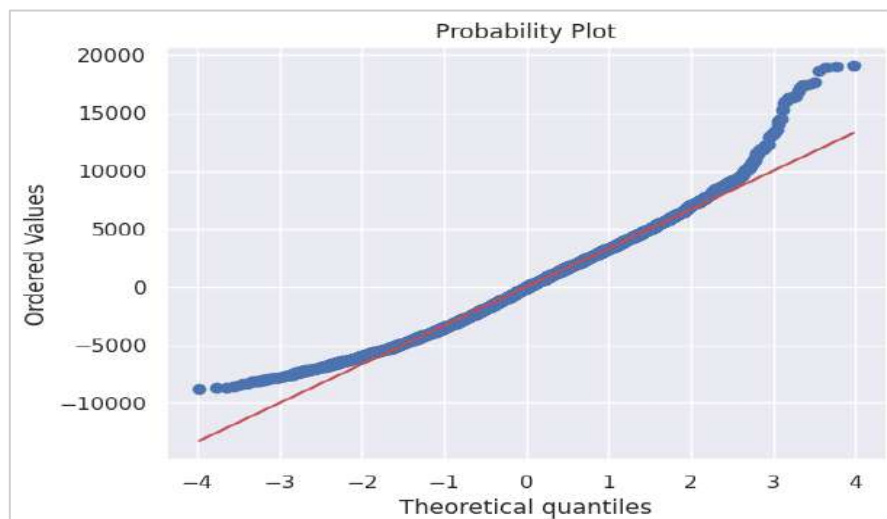


Figure 14. Q-Q Plot-Refined Baseline Model

- **Q-Q Plot:** The data points mostly followed the reference diagonal line, indicating that the residuals are **approximately normally distributed**. Minor skewness is visible at the extremes (both lower and upper tails), which is common with large sample sizes ($n = 20,000$).
- **Shapiro-Wilk test :** The **test statistic is 0.9896, p-value < 0.05 ($\sim 2.38 \times 10^{-35}$)**. Formally rejects the normality but with sample of 20,000 observations, even slight deviations reach significance. It is **practically acceptable** as it does not compromise predictive performance.

Test for Homoscedasticity

- **Goldfeld-Quandt test :** F-statistic = 0.9944, p-value = 0.6103 from the test results. Since p-value > 0.05, so we cannot reject the null hypothesis of homoscedasticity. This means that the residuals have the constant variance over the range of fitted values. The residual vs fitted value plot also showed residuals randomly distributed around zero, with no noticeable funnel shape or systematic increase/decrease in distribution, indicating constant variance.
- **Conclusion:** The Goldfeld-Quandt test and visual inspection from residual-fitted plot demonstrate that the **homoscedasticity assumption is satisfied**. Thus, no transformation is required to account for heteroscedasticity.

6.3 Final Baseline Model & Performance Evaluation

Methodology: The final baseline linear regression model was build using 9 significant features and 60:20:20 train-validation-test split, yielding 15,000 rows for training and 5,000 rows each for validation and testing. The major metric (RMSE) and supporting metrics (MAE, R-square, Adjusted R^2 and MAPE) will used to assess model performance. To preserve the Interpretability of coefficients, all models are trained using the original scaled data. All model related outputs are present in Appendix for the reference.

Final Baseline Model Performance Evaluation

=== Final Baseline Linear Regression (Training Set) ===					
RMSE	MAE	R-squared	Adj. R-squared	MAPE (%)	
3347.506313	2700.838737	0.945361	0.945329	15.028213	
=== Final Baseline Linear Regression (Validation Set) ===					
RMSE	MAE	R-squared	Adj. R-squared	MAPE (%)	
3354.315955	2668.829456	0.944874	0.944774	15.189007	

Table 12. Final baseline model (60:20:20)

- **RMSE: 3347.51 (train) and 3354.32 (validation)**, suggests model performs consistently on both the training and validation sets, also shows minimal gap indicating **no overfitting**.
- **$R^2 = 0.945$** are similar on both sets,. demonstrating that the **model generalises well on validation data**.
- Overall, the baseline linear regression model has **excellent predictive performance and stability**. Therefore, baseline benchmark established for building advanced models.

7 ADVANCED MODEL BUILDING

For the insurance cost predictions, **5 advanced regression models were chosen: Decision Tree, Random Forest, Gradient Boosting, XGBoost, and AdaBoost**. These models were chosen because the target variable might be influenced by complex relationships and interaction effects between predictors such as age, weight, and medical history. All the model's performance was evaluated on train & validation sets only to ensure test set remains unseen for the final predictions.

7.1 Decision Tree Regressor

Methodology: The Decision Tree splits the data recursively based on features with the lowest mean squared error. Predictions are made based on the average value in each leaf node.

Relevance: Captures non-linear interactions between the predictors & target variable, and gives an interpretable baseline for tree-based models.

DTR Training Set Performance:					
RMSE	MAE	R-squared	Adj. R-squared	MAPE (%)	
133.287234	6.416800	0.999913	0.999913	0.042278	
DTR Validation Set Performance:					
RMSE	MAE	R-squared	Adj. R-squared	MAPE (%)	
4212.695993	3258.253600	0.913050	0.912893	15.441546	

Table 13. DTR Performance Evaluation

- **Train set:** $R^2 \approx 0.9999$, $RMSE = 133$, showing near-perfect memorisation of the training data, suggesting **severe overfitting**.
- **Validation set:** $R^2 = 0.913$, $RMSE \approx 4213$, performance is significantly lower, indicating **poor generalisation**.
- This concludes, while the decision tree model captures complex patterns, but it **lacks generalisability and reliable predictive performance on new data** and model is not viable without further tuning.

7.2 Random Forest Regressor

Methodology: Random Forest builds multiple decision trees using bootstrap samples and averages their predictions to increase performance and reduce variance.

Relevance: Reduces overfitting while properly handling feature interactions.

Random Forest Training Set Performance:					
RMSE	MAE	R-squared	Adj. R-squared	MAPE (%)	
1179.477492	924.992631	0.993217	0.993213	4.517198	
Random Forest Validation Set Performance:					
RMSE	MAE	R-squared	Adj. R-squared	MAPE (%)	
3083.159635	2455.617697	0.953426	0.953342	11.949105	

Table 14. Random Forest Performance Evaluation

- **Train set:** $R^2 = 0.993$, $RMSE \approx 1179$, demonstrate model's **robust learning** ability without extreme memorisation.
- **Validation set:** $R^2 = 0.953$, $RMSE \approx 3083$, suggests model's consistently strong performance and **robust generalisation** to new data, **controlled overfitting as compared to the Decision Tree model**.

- The validation set's relatively **low RMSE and MAE** suggest **good prediction accuracy**, making Random Forest a reliable model for this situation.
- Overall, the model strikes an appropriate balance between bias and variance, giving it a great contender for additional tuning and final model selection.

7.3 Gradient Boosting Regressor

Methodology: Gradient Boosting builds trees sequentially, with each new tree correcting the errors of the previous model.

Relevance: Increases prediction accuracy by focusing on residual errors.

Gradient Boosting Regressor Training Set Performance:					
RMSE	MAE	R-squared	Adj. R-squared	MAPE (%)	
2940.551896	2365.906663	0.957839	0.957813	11.652507	
Gradient Boosting Regressor Validation Set Performance:					
RMSE	MAE	R-squared	Adj. R-squared	MAPE (%)	
2932.900850	2357.620479	0.957855	0.957779	11.720966	

Table 15. GB Model Performance Evaluation

- **Train RMSE = 2941 and validation RMSE ≈ 2933**, nearly identical, suggesting **minimal overfitting**.
- The nearly comparable **R² values (≈ 0.958)**, indicates **strong generalisation**
- Error measures (**RMSE, MAE, MAPE**) **are stable**, indicating that the model adequately captures underlying patterns without memorising data.
- Overall, Gradient Boosting has a great balance of bias and variance, making it a viable choice for further tuning.

7.4 XGBoost Regressor

Methodology: XGBoost is an optimised gradient boosting technique that incorporates regularisation and efficient computation.

Relevance: Provides excellent performance and effectively handles complex patterns.

XGBoost Training Set Performance:					
RMSE	MAE	R-squared	Adj. R-squared	MAPE (%)	
2689.181195	2144.352051	0.964739	0.964718	10.502566	
XGBoost Validation Set Performance:					
RMSE	MAE	R-squared	Adj. R-squared	MAPE (%)	
2927.765701	2342.543701	0.958003	0.957927	11.499298	

Table 16. XGB Model Performance Evaluation

- **Train: RMSE ≈ 2689, R² ≈ 0.965**, stronger learning capabilities than Gradient Boosting.
- **Validation: RMSE = 2928, R² ≈ 0.958**, indicating **robust generalisation**.
- A small rise in error from training to validation indicates **mild overfitting** within acceptable boundaries.
- The relatively **lower validation RMSE and MAE** among models suggest **high prediction accuracy**, making XGBoost a strong contender for tuning and final selection.

7.5 AdaBoost Regressor

Methodology: AdaBoost trains sequential weak learners, emphasising weight to mis-predicted observations.

Relevance: AdaBoost focuses on difficult cases and offers an alternative boosting approach.

Adaboost Training Set Performance:					
RMSE	MAE	R-squared	Adj. R-squared	MAPE (%)	
3274.235103	2712.914785	0.947727	0.947696	15.753448	
Adaboost Validation Set Performance:					
RMSE	MAE	R-squared	Adj. R-squared	MAPE (%)	
3256.976464	2696.569060	0.948027	0.947933	16.011246	

Table 17. AdaBoost Model Performance Evaluation

- **Train RMSE = 3274 / Validation RMSE $\approx$ 3257**, suggest model has **moderate performance**.
- Minimal training-validation gap, suggests little variation and **minimal overfitting**, but it also implies **poor learning ability**.
- AdaBoost produces **higher error values** than other ensemble algorithms, suggesting **weaker predictive power**.
- Even though, the model is reliable, but it does not capture complex patterns as well as Gradient Boosting or XGBoost.

Overall Insights from Advanced Model Performances

- Validation RMSE reveals that **XGBoost** and **Gradient Boosting** outperform **Random Forest**. These models maintain **high R^2 values**, indicating great predictive power and good generalisation.
- **AdaBoost** performs **moderately** but with higher error, whereas **Decision Tree** has the **lowest validation performance** due to overfitting.
- Therefore, **XGBoost, Gradient Boosting, and Random Forest were chosen for hyperparameter tuning** due to their superior validation performance and their strong ability to capture complex patterns in the data, making them well-suited for accurate insurance cost prediction.

8 ADVANCED MODEL HYPERPARAMETER TUNING

Methodology: After comparing the advanced models with the default hyperparameters, the top 3 models (**XGBoost, Gradient Boosting, and Random Forest**) were chosen for further optimisation based on validation performance (RMSE being the primary metric). Training performance was also assessed to guarantee that the models did not suffer from excessive overfitting.

GridSearchCV was used for hyperparameter tuning to systematically evaluate all parameter combinations and ensure optimal model performance. While **RandomizedSearchCV** is widely used in large-scale industrial settings for efficiency, **GridSearchCV** was chosen for this project because to its manageable parameter space and the need for a thorough and interpretable evaluation.

7.1 Random Forest Regressor Model Tuning

Random Forest Regressor model is selected for tuning due to its strong validation performance and good generalisation. Hyperparameter tuning can further improve model's stability and prediction accuracy.

```

---Hyperparameter Tuning: Random Forest---
Best Parameters: {'max_depth': 10, 'min_samples_split': 5, 'n_estimators': 200}

RF Tuned -Train Performance:
      RMSE      MAE  R-squared  Adj. R-squared  MAPE (%)
2549.307751 2030.964965  0.968312      0.968293  9.915369

RF Tuned-Validation Performance:
      RMSE      MAE  R-squared  Adj. R-squared  MAPE (%)
2965.314706 2368.909351  0.956918      0.956841 11.529327

```

Table 18. RF Tuned model performance Evaluation

- **Best parameters: max depth = 10, min samples split = 5 and n_estimators = 200.**
- The tuned random forest model **outperforms** the untuned version in terms of generalisation and has a **reduced RMSE** on the validation set (**~2965**).
- There is a moderate gap between training (**RMSE = 2549**) and validation performance, showing **controlled overfitting** after tuning.
- The model's **high R^2 (~0.957)** and **low MAPE (~11.53%)** demonstrate its ability to accurately identify underlying trends effectively and provides **reliable performance**.

7.2 Gradient Boosting Regressor Model Tuning

Gradient Boosting Regressor has shown highly consistent train and validation performance, suggesting strong generalisation. Hyperparameter tuning can enhance predictive power while controlling overfitting.

```

-----Hyperparameter Tuning: Gradient Boosting-----
Best Parameters: {'learning_rate': 0.05, 'max_depth': 5, 'n_estimators': 100}

GB Tuned - Train Performance:
      RMSE      MAE  R-squared  Adj. R-squared  MAPE (%)
2882.652660 2314.135623  0.959483      0.959458 11.473817

GB Tuned - Validation Performance:
      RMSE      MAE  R-squared  Adj. R-squared  MAPE (%)
2916.241603 2344.545926  0.958332      0.958257 11.790326

```

Table 19. GB Tuned model performance Evaluation

- **Best parameters: 'learning_rate': 0.05, 'max_depth': 5, 'n_estimators': 100 .**
- The tuned gradient boosting model's **performance is very consistent** across **training (RMSE = 2883)** and **validation (RMSE ~ 2916)**, showing **high generalisation**.
- The slight discrepancy across all metrics indicates that tuning has helped establish a **well-balanced bias-variance trade-off**.
- This model's high **R^2 (~0.958)** and **stable error metrics** make it **robust** and **reliable for insurance cost prediction tasks**.

7.3 XGBoost Regressor Model Tuning

XGBoost Regressor has achieved the best validation performance among all models, making it a good candidate for the final model selection. Thus, tuning is necessary to optimise predictive accuracy and ensure model generalisation.

```

-----Hyperparameter Tuning: XGBoost-----
Best Parameters: {'learning_rate': 0.05, 'max_depth': 5, 'n_estimators': 200}

XGB - Train Performance:
      RMSE      MAE  R-squared  Adj. R-squared  MAPE (%)
2810.052846 2250.689453  0.961498      0.961475 11.026328

XGB - Validation Performance:
      RMSE      MAE  R-squared  Adj. R-squared  MAPE (%)
2918.313211 2341.372070  0.958273      0.958198 11.541804

```

Table 20. XGB Tuned model performance Evaluation

- The best parameters for XGBoosting are '**learning_rate**': 0.05, '**max_depth**': 5, '**n_estimators**': 200
- The tuned XGBoost model yields **high prediction accuracy**, with a **validation RMSE** = ~2918, comparable to Gradient Boosting.
- The training-validation gap is relatively small, indicating strong generalisation and **minimal overfitting**.
- The slightly **lower MAE and MAPE** compared to other models demonstrate the ability to offer more **precise predictions** on average.

9 MODEL PERFORMANCE COMPARISON & FINAL MODEL SELECTION

8.1 Model Comparison

Methodology: All models by comparing the train and validation results across the key metrics (RMSE) as well as supporting metrics (MAE, R-squared, Adjusted R-squared, MAPE). While RMSE is used as the primary criterion for model comparison, the final model is selected based on a balanced evaluation of all metrics to ensure both accuracy and generalisability.

=====Training Performance Comparison of All Models=====						
Model	RMSE	MAE	R-squared	Adj. R-squared	MAPE (%)	
Decision Tree	133.287234	6.416800	0.999913	0.999913	0.042278	
Random Forest	1179.477492	924.992631	0.993217	0.993213	4.517198	
Random Forest (Tuned)	2549.307751	2030.964965	0.968312	0.968293	9.915369	
XGBoost	2689.181195	2144.352051	0.964739	0.964718	10.502566	
XGBoost (Tuned)	2810.052846	2250.689453	0.961498	0.961475	11.026328	
Gradient Boosting (Tuned)	2882.652660	2314.135623	0.959483	0.959458	11.473817	
Gradient Boosting	2940.551896	2365.906663	0.957839	0.957813	11.652507	
AdaBoost	3274.235103	2712.914785	0.947727	0.947696	15.753448	
Linear Regression (Baseline)	3347.506313	2700.838737	0.945361	0.945329	15.028213	
=====Validation Performance Comparison of All Models=====						
Model	RMSE	MAE	R-squared	Adj. R-squared	MAPE (%)	
Gradient Boosting (Tuned)	2916.241603	2344.545926	0.958332	0.958257	11.790326	
XGBoost (Tuned)	2918.313211	2341.372070	0.958273	0.958198	11.541804	
XGBoost	2927.765701	2342.543701	0.958003	0.957927	11.499298	
Gradient Boosting	2932.900850	2357.620479	0.957855	0.957779	11.720966	
Random Forest (Tuned)	2965.314706	2368.909351	0.956918	0.956841	11.529327	
Random Forest	3083.159635	2455.617697	0.953426	0.953342	11.949105	
AdaBoost	3256.976464	2696.569060	0.948027	0.947933	16.011246	
Linear Regression (Baseline)	3354.315955	2668.829456	0.944874	0.944774	15.189007	
Decision Tree	4212.695993	3258.253600	0.913050	0.912893	15.441546	

Table 21. Model Performance Comparison

- The comparison results show that **ensemble-based models frequently outperform the baseline linear regression model**, emphasising the presence of nonlinear relationships in the data.
- Boosting techniques outperform all other models in terms of predictive performance, with extremely close RMSE values and consistently strong supporting metrics.
- The minor difference in training and validation performance for these models indicates **high generalisation** ability.

8.2 Final Model Selection

Gradient Boosting (Tuned) is chosen as the final model because it has the lowest RMSE (2916) with high R^2 (0.958) with stable supporting metrics like MAE, R-squared, and MAPE.

Moreover, model's smallest training to validation, suggests **strong generalization** and **minimal overfitting**. While XGBoost (Tuned) has similar performance, Gradient Boosting is preferred for its slightly better RMSE and **stable results across all evaluation metrics**. Therefore, Gradient Boosting (Tuned) **offers the best mix of accuracy and stability** for this problem statement.

8.3 Final Model Performance Evaluation (Test Set)

Methodology: The final model performance evaluation was done using the main metric (RMSE) and other supporting metrics, like MAE, R-squared, Adjusted R-squared, and MAPE on an unseen test data.

-----Final Test Set Performance on Best Model (Gradient Boosting - Tuned)-----					
RMSE	MAE	R-squared	Adj. R-squared	MAPE (%)	
2977.721725	2384.647836	0.957039	0.956961	11.818321	

Table 22. Final Model Performance (Test Set)

- The final tuned Gradient Boosting model **performs well on the unseen test set**, with an **RMSE of roughly 2978 and MAE of around 2385**, indicating accurate cost predictions.
- The model explains a **significant proportion of the variance in insurance costs**, as indicated by the **high R-squared value (~0.957)**.
- The model's **strong generalisation and no overfitting** are verified by the close alignment between test and validation results, making it reliable for real-world applications.

8.4 Final Model Predictions (Test Set)

Methodology: To assess the final model's practical applicability, predictions was made on an unseen test dataset by comparing the actual and predicted insurance costs as shown below. If the predicted values are relatively close to the actual values, with moderate prediction errors. Then the model generalises effectively to new data and can produce accurate predictions for insurance cost prediction.

=== Sample Predictions (Actual vs Predicted) ===			
	Actual	Predicted	Error
0	38254	38902.270000	-648.270000
1	6170	6444.740000	-274.740000
2	17276	21029.500000	-3753.500000
3	18510	16941.750000	1568.250000
4	33318	33661.690000	-343.690000

Table 23. Final Model Prediction (Test Data)

- The **prediction values closely aligns with actual insurance costs**, with relatively small prediction errors (e.g., ± 300 -600), indicating **good accuracy** in typical scenarios.
- Some instances (e.g., ~3750) show larger errors, indicating that the model may struggle with specific circumstances, presumably due to complicated or extreme feature combinations.
- Both overestimation and underestimation are observed, but there is no consistent bias, indicating that the **model is well balanced**.
- Overall, the model has **high prediction capabilities and acceptable error margins** for real-world applications.

8.5 Feature Importance

Methodology: Since Gradient Boosting is an ensemble of decision trees, feature importance can be derived based on the contribution of each feature in reducing prediction error across all trees.

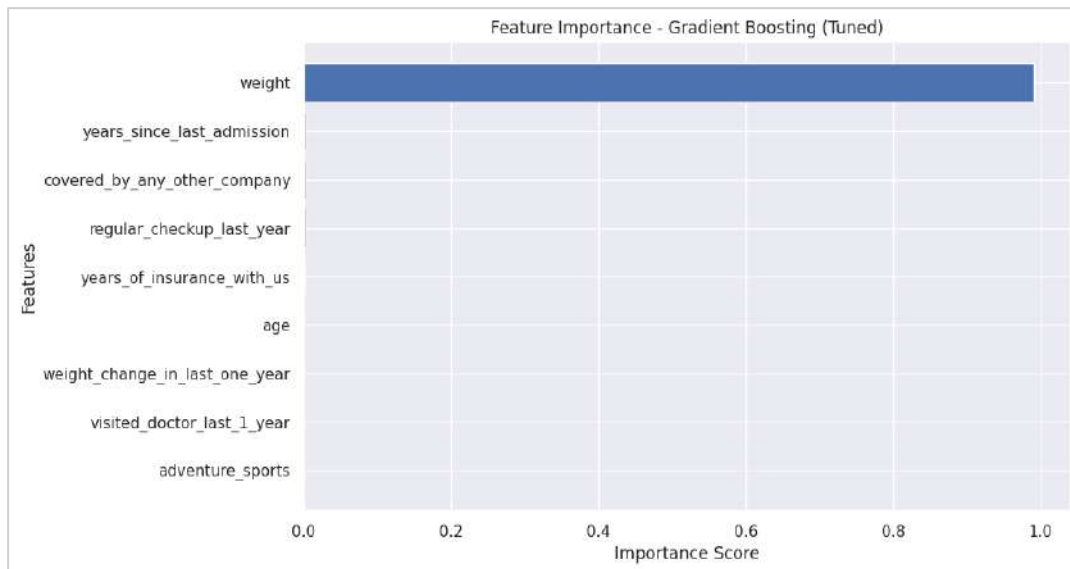


Figure 15. GB Model Feature Importance

- **Primary contributor: weight is the leading predictor**, accounting for >99% of the model's feature importance, demonstrating that an individual's weight has a significant influence on insurance costs in this dataset.
- **Secondary contributors:** years since last admission, coverage by another company, and regular health check-ups, indicate that they have a **secondary role in influencing cost**.
- The **remaining factors have an extremely low relevance**, indicating that their **impact on prediction is small** when compared to weight.
- This result is consistent with earlier EDA insights, where **weight identified as a key driver**. However, the **high dominance of a single feature** warrants monitoring. Therefore, periodic model retraining is recommended to ensure robustness.

10 ACTIONABLE INSIGHTS & BUSINESS RECOMMENDATIONS

Based on the full analytical and modeling pipeline, the following insights & recommendations directly support the business objective of accurately estimating insurance cost and optimising risk-based pricing.

1. Weight is the Dominant Factor of Insurance Cost

Insight:

- Feature importance analysis shows that weight is the most important factor when predicting insurance cost.

Recommendation:

- Since weight is closely linked to health risk, it should be included in pricing strategies. Insurance providers can encourage healthier lifestyles by offering incentives, like lower premiums for those who maintain a healthy weight. This would help reduce long-term risk.

2. Preventative Healthcare Helps Lower Costs

Insight:

- Regular check-ups and preventive health behaviours lead to lower predicted insurance costs.

Recommendation:

- Promoting preventive care can help lower claims and overall risk. Insurance companies can create policies that reward people who actively manage their health, which would improve customer outcomes and cut costs.

3. Medical History Shows Risk Exposure

Insight:

- Factors related to past hospital visits indicate that recent medical events greatly affect insurance costs.

Recommendation:

- Including medical history in pricing models allows for more accurate risk-based pricing, ensuring that premiums match the true risk profile of individuals.

4. Customer Behaviour and Lifestyle Impact Risk

Insight:

- Health and lifestyle factors together influence predicted insurance costs, highlighting the importance of individual behaviour in assessing risk.

Recommendation:

- Insurance companies should consider behavioral factors in underwriting decisions and promote healthier habits through specific programs, which could lower the chances of high-cost claims.

5. Model Supports Data-Driven Pricing Decisions

Insight:

- The final model provides reliable predictions of insurance costs based on health and lifestyle data.

Recommendation:

- This model can serve as a decision-making tool to optimize premium pricing, enhance underwriting accuracy, and categorize customers based on risk levels.

6. Monitor Model Dependence on Weight

Insight:

- The model depends heavily on weight as the primary predictor.

Recommendation:

- While this boosts prediction accuracy, insurers should track model performance to ensure reliability across different customer groups and avoid relying too much on a single factor.

7. Continuous Model Monitoring is Crucial for Long-Term Use

Insight:

- Changes in customer health patterns and external factors can influence model performance over time.

Recommendation:

- Regular retraining and validation of the model will ensure that it continues to deliver accurate and trustworthy cost predictions in a dynamic business environment.

11 APPENDIX

This appendix provides supporting material including extended EDA plots, raw statistical outputs, extended data processing outputs, and full model summaries referenced from the main report.

11.1 Extended Data Overview

The following tables covers all the raw outputs which are not shown in the main body (Section 2).

Checking the data types

There are 8 categorical variables and 16 numerical variables.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25000 entries, 0 to 24999
Data columns (total 24 columns):
 #   Column                                Non-Null Count  Dtype  
---  --
 0   applicant_id                         25000 non-null  int64  
 1   years_of_insurance_with_us          25000 non-null  int64  
 2   regular_checkup_last_year           25000 non-null  int64  
 3   adventure_sports                     25000 non-null  int64  
 4   Occupation                           25000 non-null  object  
 5   visited_doctor_last_1_year          25000 non-null  int64  
 6   cholesterol_level                   25000 non-null  object  
 7   daily_avg_steps                     25000 non-null  int64  
 8   age                                 25000 non-null  int64  
 9   heart_decs_history                  25000 non-null  int64  
10   other_major_decs_history             25000 non-null  int64  
11   Gender                               25000 non-null  object  
12   avg_glucose_level                   25000 non-null  int64  
13   bmi                                  24010 non-null  float64 
14   smoking_status                       25000 non-null  object  
15   Year_last_admitted                  13119 non-null  float64 
16   Location                             25000 non-null  object  
17   weight                               25000 non-null  int64  
18   covered_by_any_other_company         25000 non-null  object  
19   Alcohol                             25000 non-null  object  
20   exercise                             25000 non-null  object  
21   weight_change_in_last_one_year       25000 non-null  int64  
22   fat_percentage                       25000 non-null  int64  
23   insurance_cost                       25000 non-null  int64  
dtypes: float64(2), int64(14), object(8)
```

Table 24. Appendix- Data types

Converting Data Types

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25000 entries, 0 to 24999
Data columns (total 24 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   applicant_id                          25000 non-null  int64
1   years_of_insurance_with_us            25000 non-null  int64
2   regular_checkup_last_year             25000 non-null  int64
3   adventure_sports                       25000 non-null  int64
4   Occupation                             25000 non-null  category
5   visited_doctor_last_1_year             25000 non-null  int64
6   cholesterol_level                      25000 non-null  category
7   daily_avg_steps                        25000 non-null  int64
8   age                                    25000 non-null  int64
9   heart_decs_history                     25000 non-null  int64
10  other_major_decs_history                25000 non-null  int64
11  Gender                                  25000 non-null  category
12  avg_glucose_level                      25000 non-null  int64
13  bmi                                    24010 non-null  float64
14  smoking_status                          25000 non-null  category
15  Year_last_admitted                     13119 non-null  float64
16  Location                                25000 non-null  category
17  weight                                 25000 non-null  int64
18  covered_by_any_other_company            25000 non-null  category
19  Alcohol                                 25000 non-null  category
20  exercise                                25000 non-null  category
21  weight_change_in_last_one_year          25000 non-null  int64
22  fat_percentage                          25000 non-null  int64
23  insurance_cost                          25000 non-null  int64
dtypes: category(8), float64(2), int64(14)
memory usage: 3.2 MB
```

Table 25. Appendix- Converted data types

- We can observe that the memory usage has decreased from 4.6+ MB to 3.2 MB, this technique is generally useful for bigger datasets.

Checking unique value & counts

- Top region is Bangalore.
- Top occupation found to be student.
- There are more male customers as compared to female.

```
Unique values in Location are :
Location
Bangalore    1742
Jaipur        706
Bhubaneswar  1784
Mangalore    1697
Delhi        1680
Ahmedabad    1677
Guwahati     1672
Chennai      1669
Kanpur       1664
Nagpur       1663
Mumbai       1658
Lucknow      1637
Pune         1622
Kolkata      1620
Surat        1589
Name: count, dtype: int64
*****
Unique values in covered_by_any_other_company are :
covered_by_any_other_company
N      17418
Y       7582
Name: count, dtype: int64
*****
Unique values in Alcohol are :
Alcohol
Rare    13752
No       8541
Daily   2707
Name: count, dtype: int64
*****
Unique values in exercise are :
exercise
Moderate  14638
Extreme   5248
No         5114
Name: count, dtype: int64
*****

Unique values in Occupation are :
Occupation
Student    10169
Business   10020
Salried     4811
Name: count, dtype: int64
*****
Unique values in cholesterol_level are :
cholesterol_level
150 to 175    8763
125 to 150    8339
200 to 225    2963
175 to 200    2801
225 to 250    2054
Name: count, dtype: int64
*****
Unique values in Gender are :
Gender
Male    16422
Female   8578
Name: count, dtype: int64
*****
Unique values in smoking_status are :
smoking_status
never smoked    9249
Unknown         7555
formerly smoked  4329
smokes          3867
Name: count, dtype: int64
*****
```

	count	unique	top	freq
Occupation	25000	3	Student	10169
cholesterol_level	25000	5	150 to 175	8763
Gender	25000	2	Male	16422
smoking_status	25000	4	never smoked	9249
Location	25000	15	Bangalore	1742
covered_by_any_other_company	25000	2	N	17418
Alcohol	25000	3	Rare	13752
exercise	25000	3	Moderate	14638

Table 26. Appendix- Unique value & counts

Dropping unnecessary variables

#	Column	Non-Null Count	Dtype
0	years_of_insurance_with_us	25000 non-null	int64
1	regular_checkup_lasy_year	25000 non-null	int64
2	adventure_sports	25000 non-null	int64
3	Occupation	25000 non-null	category
4	visited_doctor_last_1_year	25000 non-null	int64
5	cholesterol_level	25000 non-null	category
6	daily_avg_steps	25000 non-null	int64
7	age	25000 non-null	int64
8	heart_decs_history	25000 non-null	int64
9	other_major_decs_history	25000 non-null	int64
10	Gender	25000 non-null	category
11	avg_glucose_level	25000 non-null	int64
12	bmi	24010 non-null	float64
13	smoking_status	25000 non-null	category
14	Year_last_admitted	13119 non-null	float64
15	Location	25000 non-null	category
16	weight	25000 non-null	int64
17	covered_by_any_other_company	25000 non-null	category
18	Alcohol	25000 non-null	category
19	exercise	25000 non-null	category
20	weight_change_in_last_one_year	25000 non-null	int64
21	fat_percentage	25000 non-null	int64
22	insurance_cost	25000 non-null	int64

dtypes: category(8), float64(2), int64(13)
memory usage: 3.1 MB

Table 27. Appendix- Dataset after dropping case_id

- We can observe **applicant_id** has been dropped.
- Now, we have 25000 rows and **23 columns**.

11.2 Extended Univariate Analysis

11.2.1 Univariate Analysis on Numerical Variables

fat_percentage

- The mean = ~ 28.8%, and the std dev = 8.63, Range 11%-42%; slightly left-skewed; most individuals cluster in 30% -38% range. Since there are no significant extreme values in the data.
- Since the fat percentage is directly related to health conditions, it may be considered a significant predictor of insurance costs, as the percentage may be on the higher side.

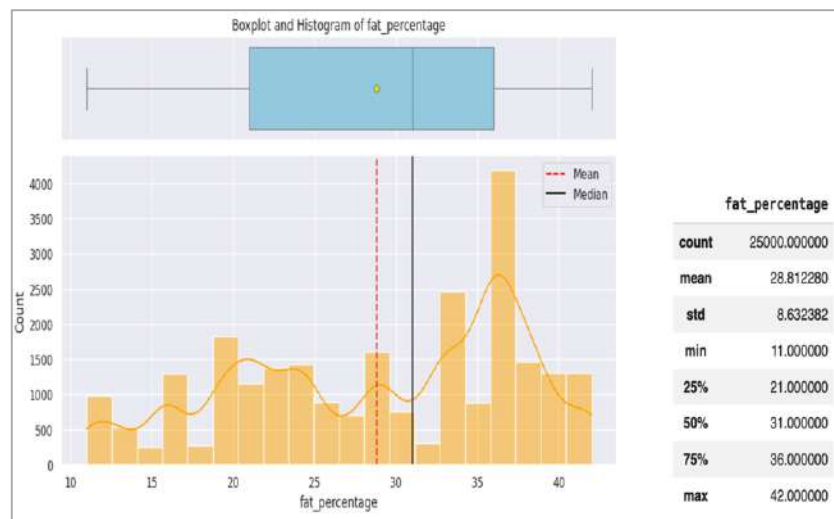


Figure 16. Appendix- Histogram-Boxplot of Fat_percentage

avg_glucose_level

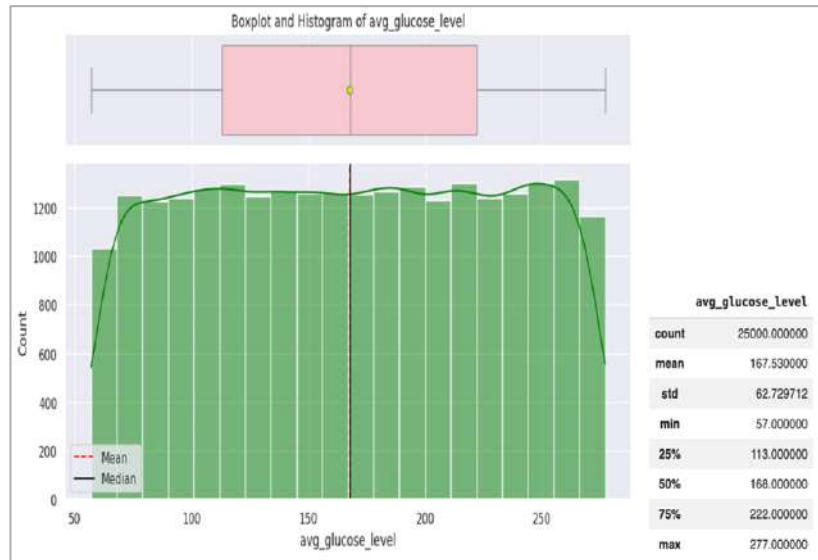


Figure 17. Appendix- Histogram-Boxplot of avg_glucose_level

- Mean ≈ 167.5 , range 57–277. Symmetric distribution; no outliers detected. Wide spread reflects a mix of normal and high-risk glucose profiles.
- This data is likely to be a good predictor for insurance costs because higher glucose levels imply a higher risk for medical complications.

daily_avg_steps

- Mean ~ 5216 ; slightly right-skew. 952 values above 7510 (outliers as per IQR), suggests genuine high activity individuals. The majority of individuals appear to fall under a moderately active category. Physical activity is inversely related to health risks; therefore, daily steps are likely to have a negative relationship with insurance cost.

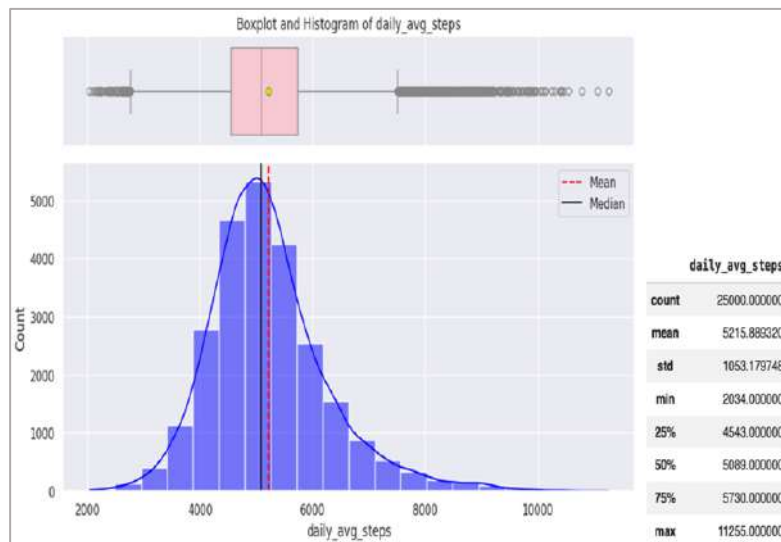


Figure 18. Appendix-Histogram-Boxplot of daily_avg_steps

BMI

- On average, the BMI is 31.39, and the standard deviation is 7.88, with range 12.30-100.60. The data is right skewed with outliers. BMI might have a positive relationship with insurance costs or medical expenditures.

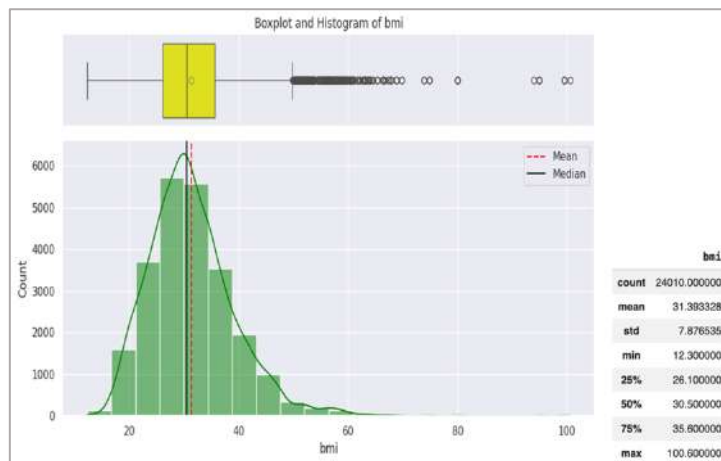


Figure 19. Appendix-Histogram-Boxplot of bmi

years_of_insurance_with_us

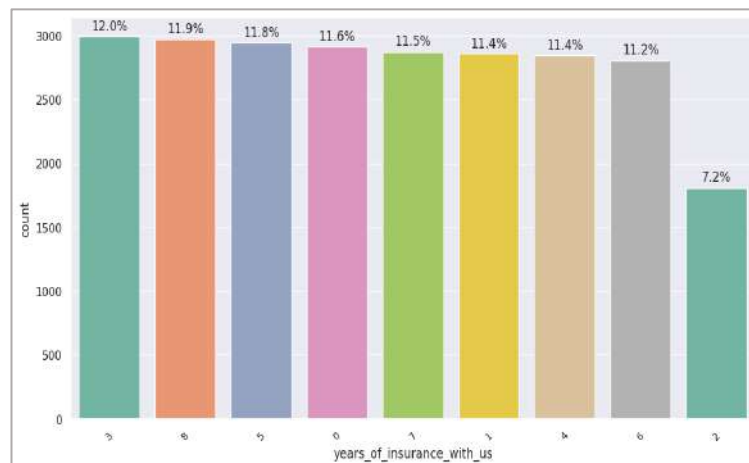


Figure 20. Appendix- Bar plot of years_of_insurance_with_us

- Mean ~31.39 (overweight/obese range); right-skewed with maximum 100.6; outliers present. It appears that the longer the insurance tenure, the more likely the customer is to be given loyalty discounts, and as such, the insurance costs are likely to have an inverse relationship with the years of insurance tenure.

Age

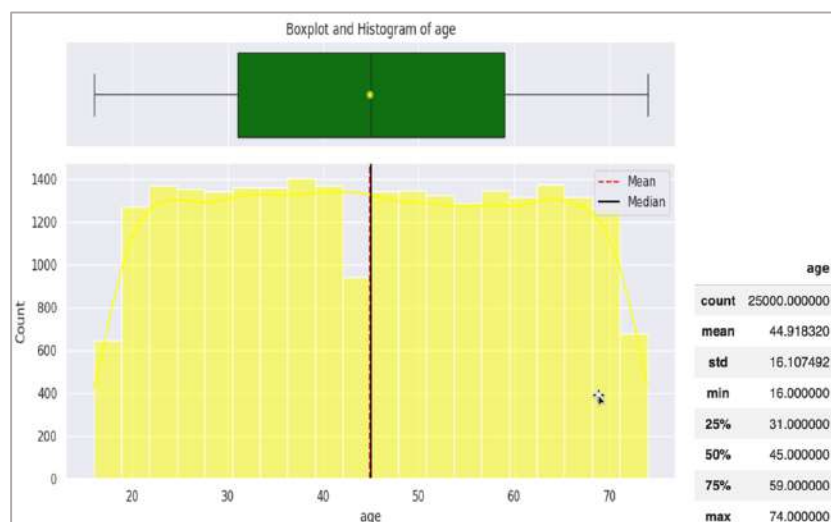


Figure 21. Appendix- Histogram-Boxplot of age

- The average age ~ 44.92 , a std dev ~ 16 , range 16- 74. The boxplot does not show any outliers. There appears to be a sudden and unnatural drop in the plot for the individuals between 42 and 45 years old. Data is uniform, this means the business has a balanced demographic for young adults, mid-life adults, and seniors.

adventure_sports

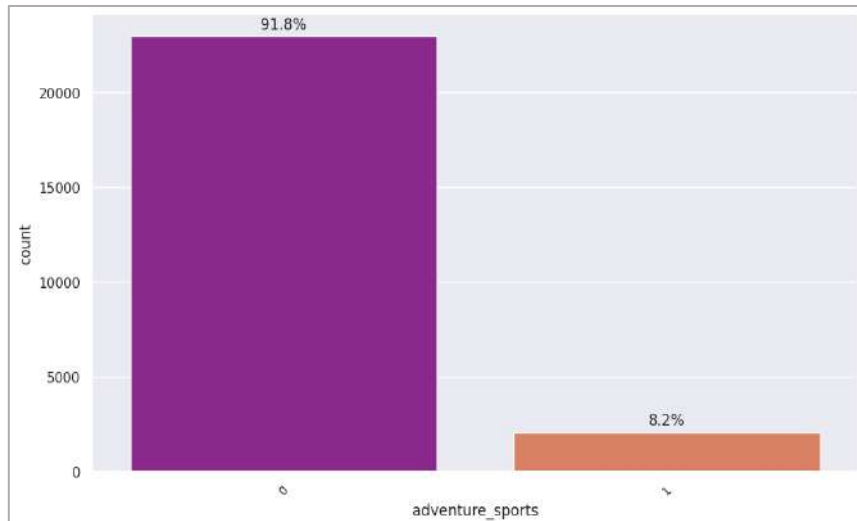


Figure 22. Appendix-Bar plot of adventure_sports

- The data type is binary, where 0 denotes the absence and 1 denotes the presence of adventure sports participation. The data is highly skewed, with 91.8% of individuals not participating in adventure sports and the remaining 8.2% participating in the same. Adventure sports participants are more likely to have a high chance of physical injury or accidents; therefore, this data is likely to have a positive correlation with insurance costs or medical expenditure claims.

visited_doctor_last_1_year

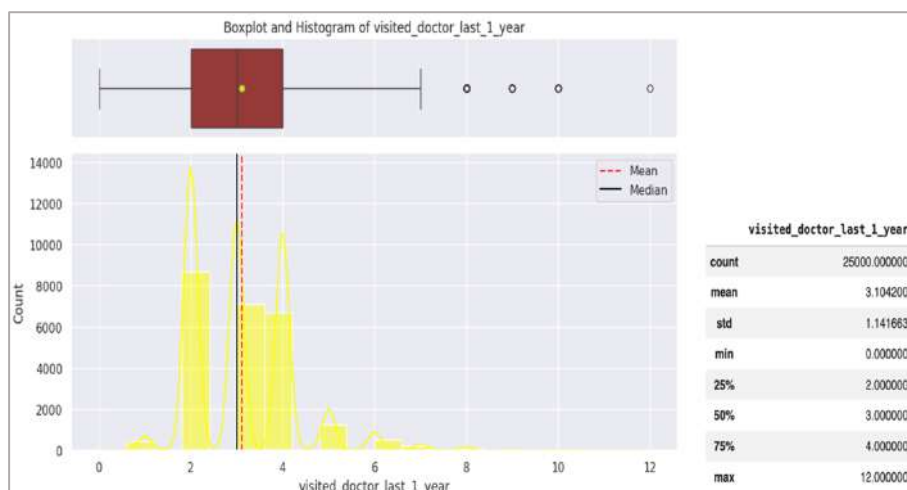


Figure 23. Appendix-Histogram-Boxplot of visited_doctor_last_1_year

- Mean ~ 3.10 ; right-skewed. outliers above 7 visits indicates as genuine high-frequency patients. This could be an indicator of an individual's underlying health status, so a high rate of visits should be positively correlated with insurance costs or medical expenditures.

heart_decs_history

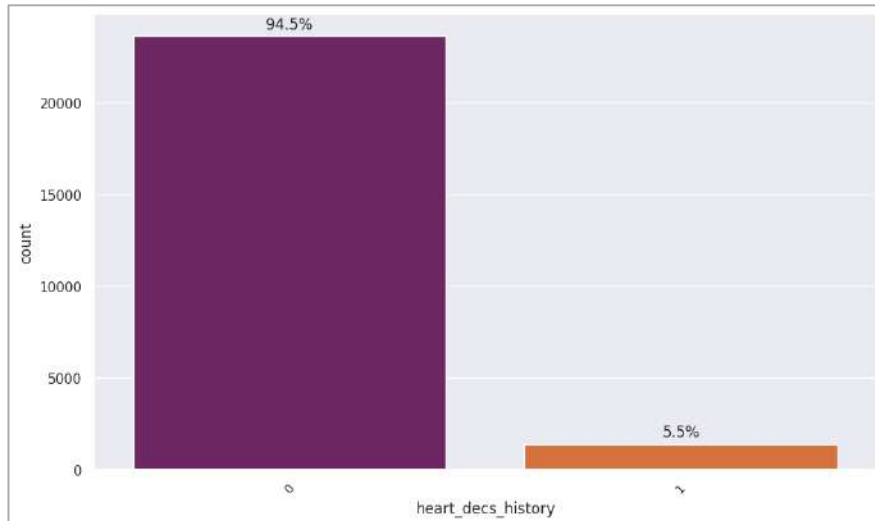


Figure 24. Appendix- Bar plot of heart_decs_history

- The data set is a binary, 0 indicates no history of heart disease and 1 indicates a history of heart disease. Data is highly skewed with 94.5% of people having no history of heart disease and only 5.5% of people having a history of heart disease. Having a history of heart disease increases the risk of medical claims and hence is related to insurance costs in a positive manner.

other_major_decs_history

- The data points are binary, 0 represents the absence and 1 represents the presence of other major disease histories, highly skewed, with 90.2% of the population showing no other major disease histories and 9.8% showing a positive history. A history of other major disease histories or surgeries generally points towards high risk levels in underwriting; hence, this factor can be expected to have a positive correlation with premium or claim costs.

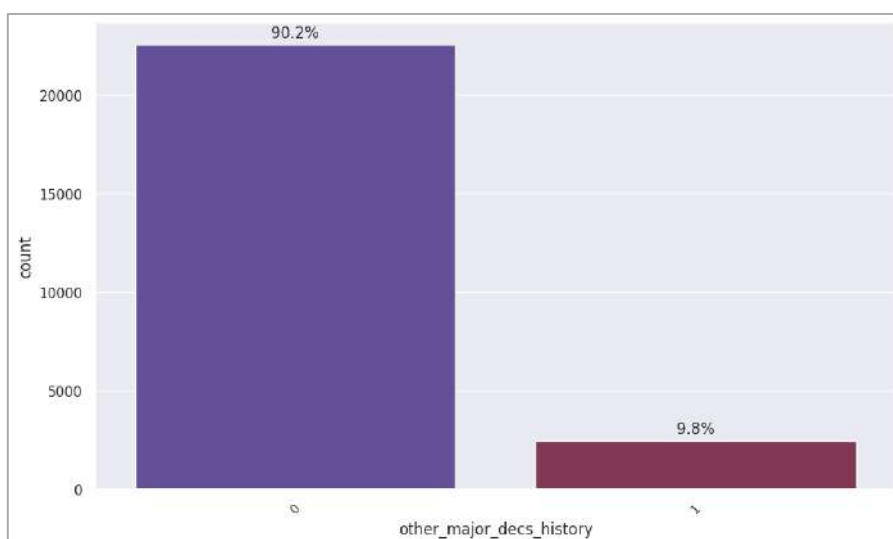


Figure 25. Appendix-Bar plot of other_major_decs_history

Year_last_admitted

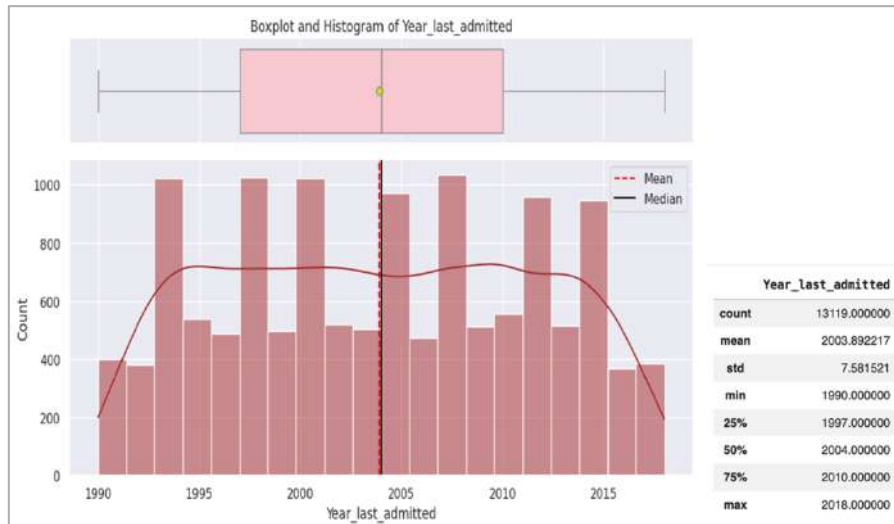


Figure 26. Appendix- Histogram-Boxplot of Year_last_admitted

- Mean ~ 2003.89 . The std dev ~ 7.58 ; Range 1990 – 2018, is uniformly distributed but multimodal. There are no mathematical outliers. The recency of hospitalization could be a significant factor in determining the premium for a patient, with higher premiums for those hospitalized closer to 2018 compared to the 1990s.

weight change in last one year

- Mean ~ 2.52 , and the standard deviation is about 1.69 units. Multi-modal; peaks only at whole numbers suggesting data rounding. Range 0–6; no outliers. A history of weight fluctuation is generally a reflection of a change in lifestyle, diet, or metabolic conditions, and this factor is likely to have a positive correlation with insurance costs.

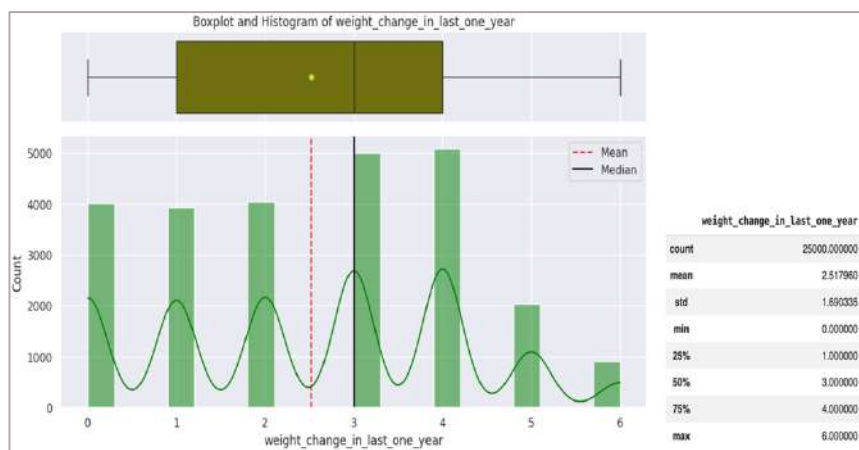


Figure 27. Appendix- Histogram- Boxplot of weight_change_in_last_one_year

11.2.2 Univariate Analysis on Categorical Variables

Categorical features were examined for class balance, data quality, and encoding suitability. Binary and ordinal mappings were applied where appropriate prior to bivariate analysis

Occupation

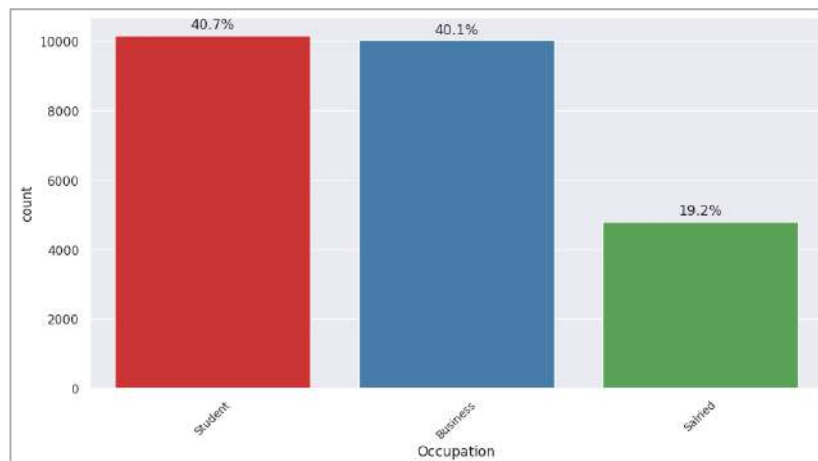


Figure 28. Appendix- Bar plot of Occupation

- The dataset representing 3 distinct career (Student (40.7%) or Business owners (40.1%), and Salaried (19.2%)). A history of an employment type, such as Business or Salaried, generally correlates with different stress profiles, different levels of physical exertion, or different income stability, which may have a direct correlation with premium or claim costs.

Cholesterol level

- The data points representing five distinct cholesterol level (125 - 150 up to 225 – 250)
- 68.5% of customers fall into the lower two bands, indicating a good cholesterol profile. Given that a history of higher levels of cholesterol (above 200) generally indicates a higher level of cardiovascular risk, this factor is likely to have a positive correlation with premium or claim costs.

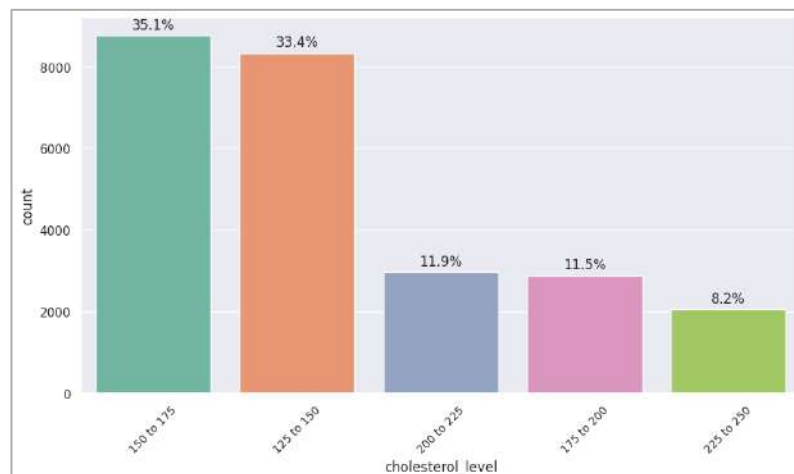


Figure 29. Appendix- Bar plot of Cholesterol_level

Gender

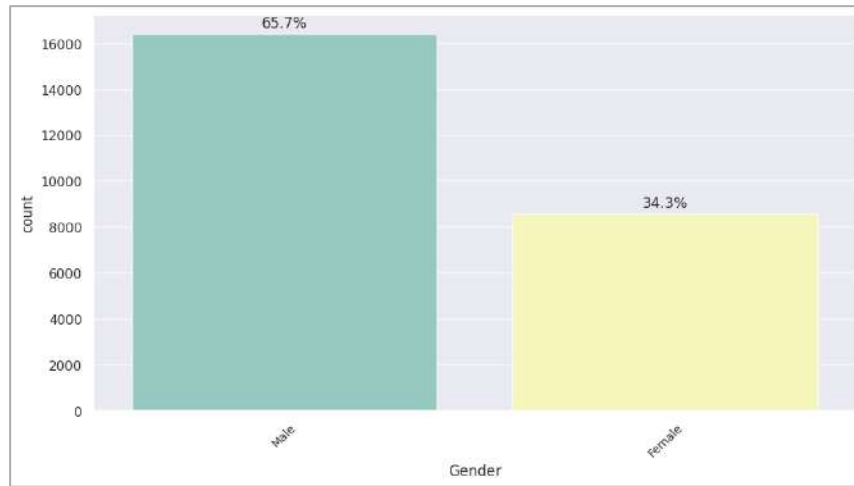


Figure 30. Appendix- Bar plot of Gender

- The data points are of binary (Male and Female), distribution is highly concentrated for Males (65.7%), where Females constitute a minority class (34.3%). The plot indicates that there is a clear class imbalance.

smoking_status

- Four categories: never smoked (37.0%), Unknown (30.2%), formerly smoked (17.3%), smokes (15.5%). "Unknown" (~30%) needs to be imputed with mode during pre-processing. From the bar plot, it is evident that there is class imbalance . The class of people who are currently smoking (smokes) indicates a higher risk of respiratory and cardiovascular risks, so it is expected that this factor would have a very high positive correlation with risk levels.

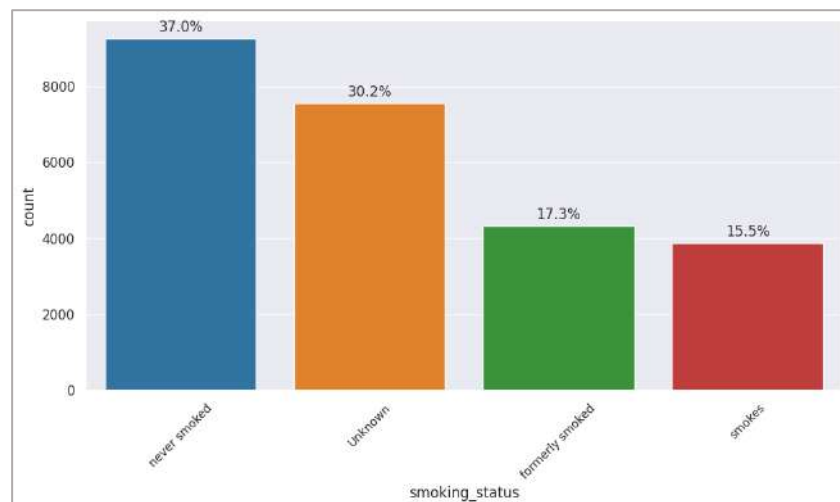


Figure 31. Appendix- Bar plot of smoking_status

Location

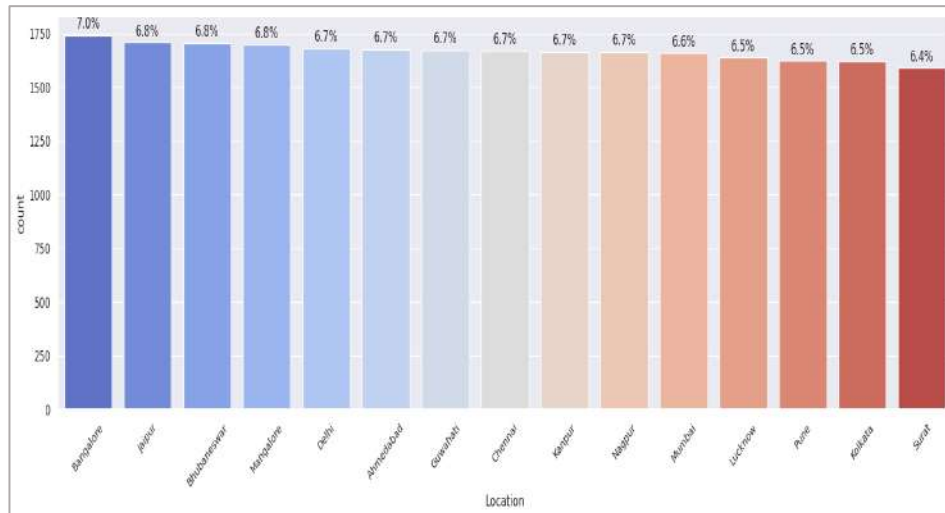


Figure 32. Appendix- Bar plot of Location

- The data points are categorical in nature, representing 15 different city-based geographical location, distribution is almost perfectly uniform, with the highest concentration being in Bangalore (7.0%) and the lowest being in Surat (6.4%). A history of geographical locations usually implies a direct correlation with regional healthcare infrastructure costs, pollution indexes, and cost-of-living adjustments in underwriting.

Alcohol

- The categories include the Rare (55.0%), No (34.2%), and Daily (10.8%) consumption brackets.
- The visual representation of the feature clearly shows a large class imbalance, as the Rare category of alcohol consumption is more than five times larger than the segment of people consuming alcohol every day. This feature should theoretically show a positive correlation with the cost of underwriting and premiums.

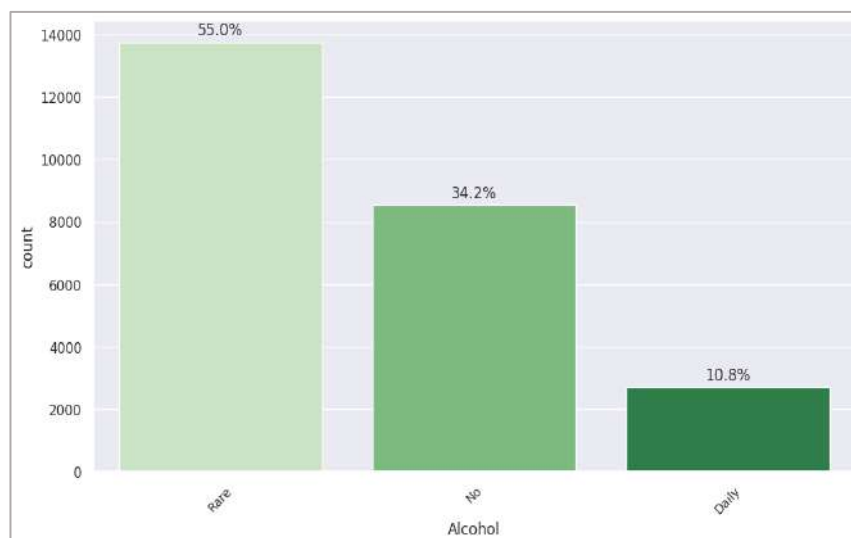


Figure 33. Appendix- Bar plot of Alcohol

Exercise

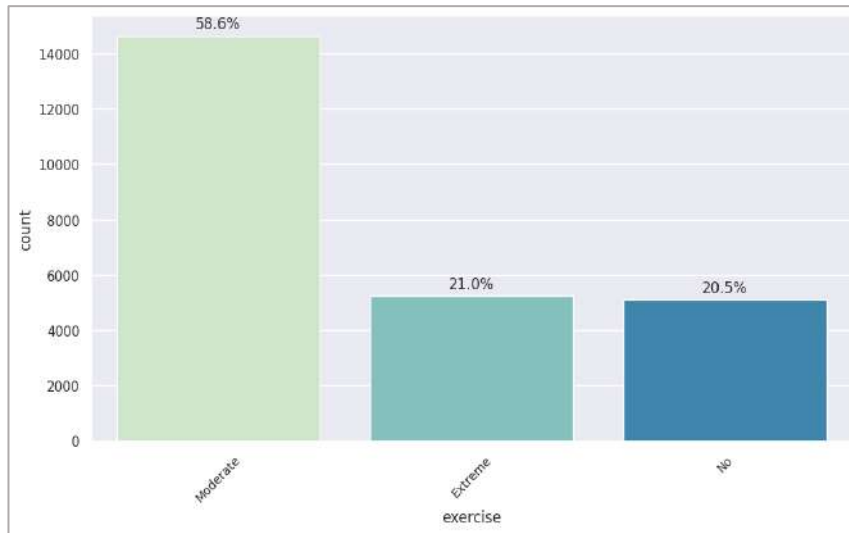


Figure 34. Appendix- Bar plot of exercise

- This feature categories are spread out along the scale of Moderate (58.6%), Extreme (21%), and No Exercise (20.5%).
- The results indicates a high level of skewness. Individuals with Sedentary Lifestyle (No Exercise) are known to be at a high risk of heart disease, and individuals with Extreme physical training are at a risk of sports injury; thus, these two categories are expected to have different correlations with respect to policy costs and premium pay-outs.

Binary Encoding of Categorical Variables before Bivariate Analysis

Before proceeding to bivariate analysis, we binary encoded (Mapping to 0 and 1) columns having for simple Yes/No, or Gender, covered_by_any_other_company columns. For columns where the categories have a natural ranking from low to high, we will use ordinal mapping: Alcohol (No = 0, Rare = 1, Daily = 2), exercise (No = 0, Moderate = 1, Extreme = 2) and cholesterol_level (125 to 150 = 0, 150 to 175 = 1, etc.).

	Gender	covered_by_any_other_company	Alcohol	exercise	cholesterol_level
0	0	0	0	1	0
1	0	0	0	1	1
2	1	0	0	2	2
3	1	1	1	1	0
4	0	0	0	0	2

Table 28. Appendix- Binary Encoding of Categorical Columns

11.3 Extended Bivariate Analysis

Extended bivariate analysis includes all feature-vs-target and feature-vs-feature connections that were investigated. These graphs guided feature selection and feature engineering strategy. Bivariate feature selection was influenced by statistical association signals from EDA, domain knowledge of healthcare risk factors, and avoiding repetition among closely related variables.

Bivariate Analysis on Target vs Numerical Features

In this step, for analysis we will use Regplot instead of Scatter plot for plotting as it instantly overlay a fitted regression line and confidence interval on the data, making trends and linear relationships clearer than points alone.

BMI vs Insurance Cost

It examines whether the composition of the human body, i.e., the relative weight to height ratio, influences insurance premiums.

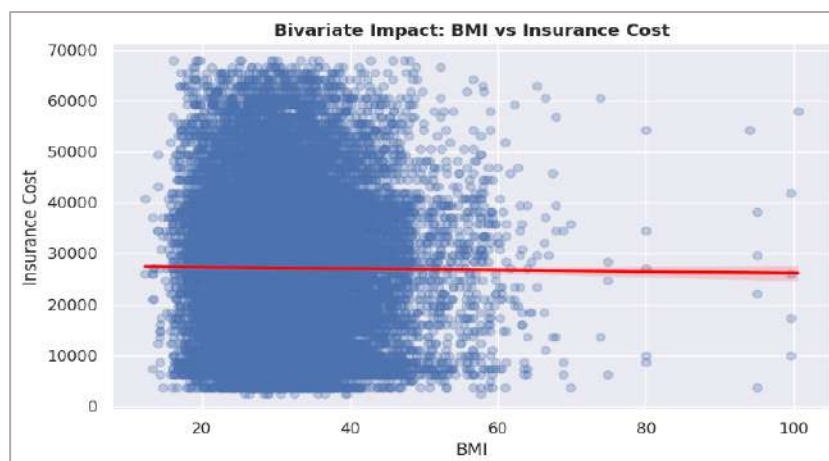


Figure 35. Appendix- Regplot of bmi vs insurance_cost

- The insurance costs are spread out across all the values of the BMI (2000 to 65000), with no particular trend upwards or downwards. The line also remains flat, showing no particular level of correlation between the insurance cost and the BMI.

Fat Percentage vs Insurance Cost

It analyses whether a high percentage of fat composition in the human body, which is a more accurate measure of health, influences insurance premiums.

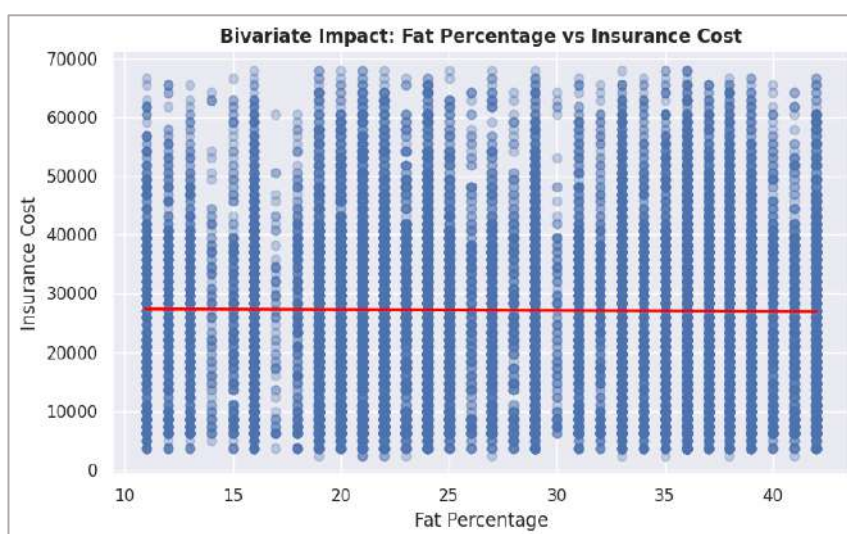


Figure 36. Appendix- Regplot of fat_percentage vs insurance_cost

- The insurance costs vary widely, ranging from 3000 to 67000. All the data points are included, but no specific pattern can be identified. Fat percentage, although a more specific measure of health, does not show any direct correlation with insurance costs.

Average Glucose Level vs Insurance Cost

It investigates the effect of average glucose levels on insurance premiums, which may be a measure of diabetes.

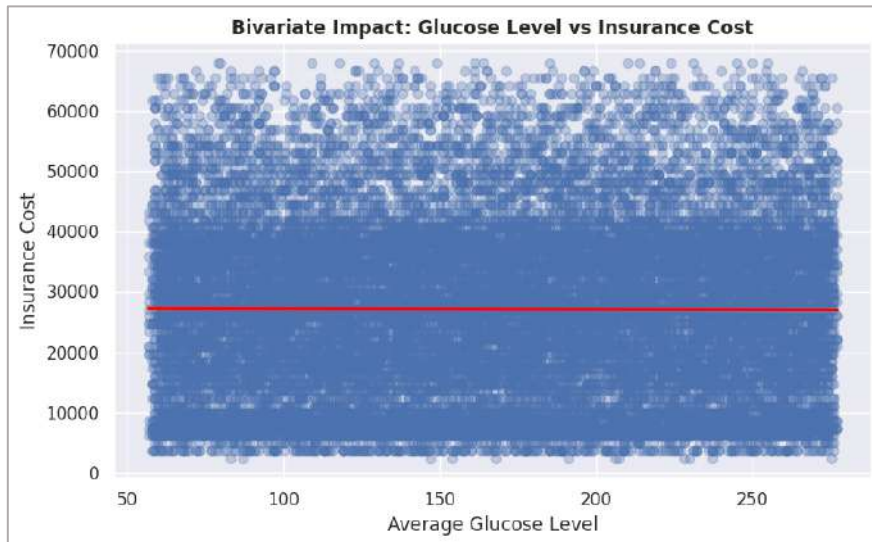


Figure 37. Appendix- Regplot of Avg_glucose_level vs insurance_cost

- Insurance costs are highly variable (2000 to 67000) for all glucose level data points, with no concentration or pattern. The regression line remains flat, indicating no linear relationship between the glucose level and insurance cost.

Average Daily Steps vs Insurance Cost

It examines whether increased physical activity levels have a direct correlation with reduced insurance premiums.

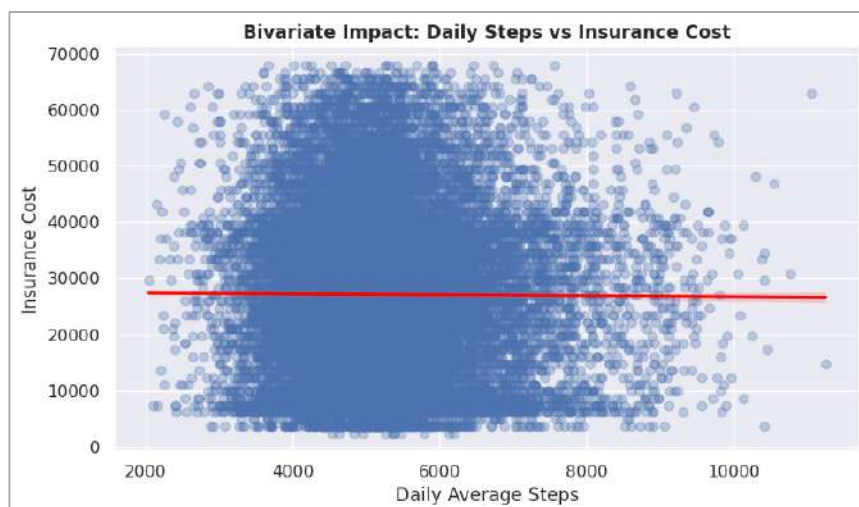


Figure 38. Appendix- Regplot of daily_avg_steps vs insurance_cost

- Insurance costs are evenly spread over the entire range of data, between 2000 and 68000, for all step levels, with no specific clustering of data points in any particular step level. The red regression line is flat and horizontal, indicating that there is no specific linear relationship between the data points.

Age vs Insurance Cost

It determine whether older age groups s have a direct correlation with increased insurance premiums.

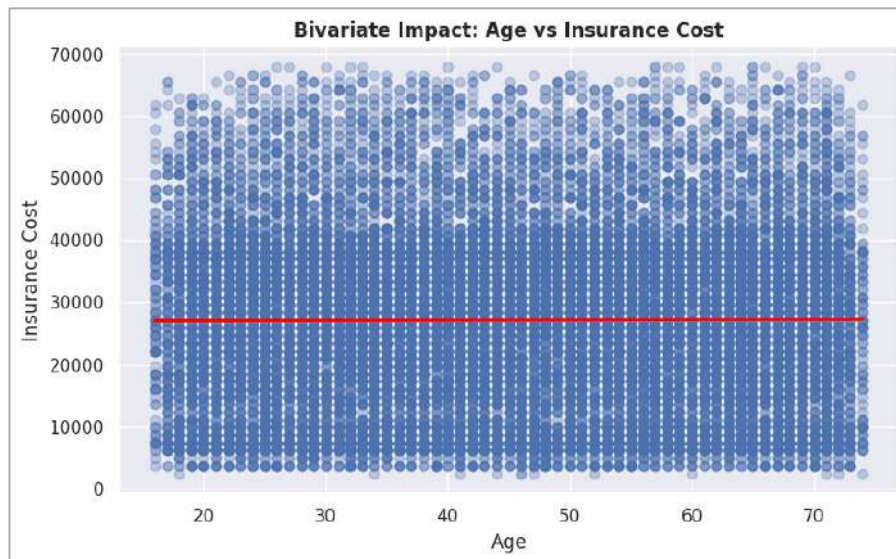


Figure 39. Appendix- Regplot of age vs insurance_cost

- Insurance costs range from less than 5,000 to almost 70,000 for every age group. There isn't an upward diagonal slope. In this dataset, age by itself has no linear predictive power over insurance costs. The maximum premium of 70,000 may be applied to both a 20-year-old and a 70-year-old. It demonstrates that other metrics (such as smoking or BMI) totally supersede age as a cost driver because costs are evenly distributed across all age groups.

Bivariate Analysis on Target vs Categorical Features

We used **Boxplots for analysis** as it efficiently compare the distribution, central tendency (median), spread, and outliers of numerical data across different categories side-by-side.

Exercise vs Insurance Cost

This plot evaluates how different levels of exercises influence insurance cost and individual health risks.

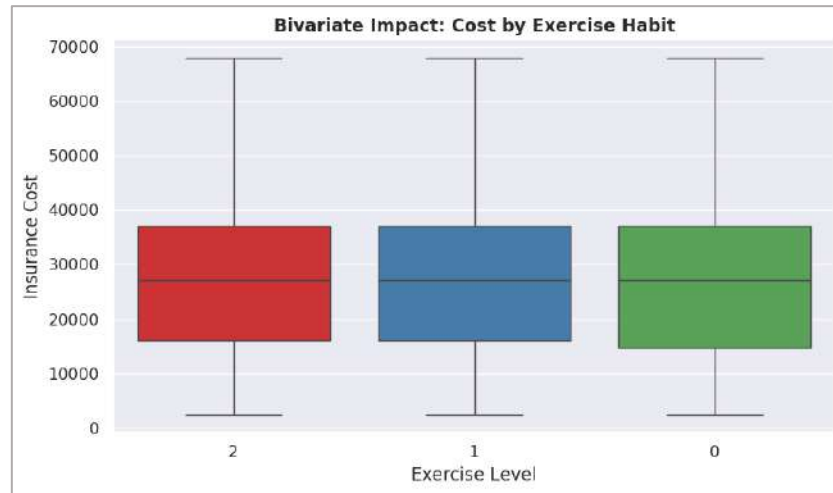


Figure 40. Appendix- Boxplot of exercise vs insurance_cost

- The cost of insurance varies greatly between roughly 3000 and 68000 for all exercise levels (0, 1, 2). When exercise level rises or falls, there is no discernible change in distribution. This suggests that exercise might not be a significant pricing factor on its own and that its impact might only become apparent when paired with other lifestyle or medical factors.

Heart Disease History vs Insurance Cost

This plot analyses whether individuals suffering from heart disease are charged more for insurance due to increased medical risks.

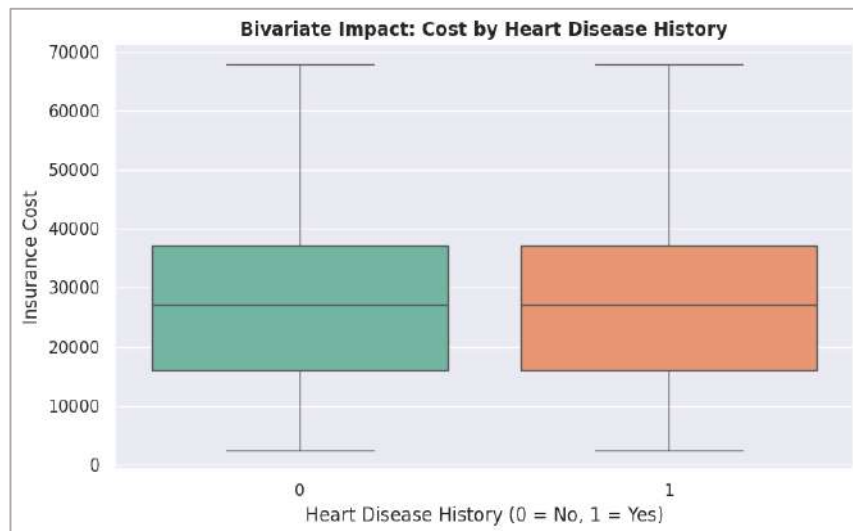


Figure 41. Appendix- Boxplot of heart_decs_history vs insurance_cost

- Both groups insurance premiums (0: No history, 1: History) fall between roughly 3000 and 68,000. For those with a history of heart disease, there is no discernible increase in expenses. Given its clinical significance, heart disease history does not appear to have a significant impact on insurance costs on its own.

Location vs Insurance Cost

This analyse if location have any significant effect on insurance pricing based on different regions.

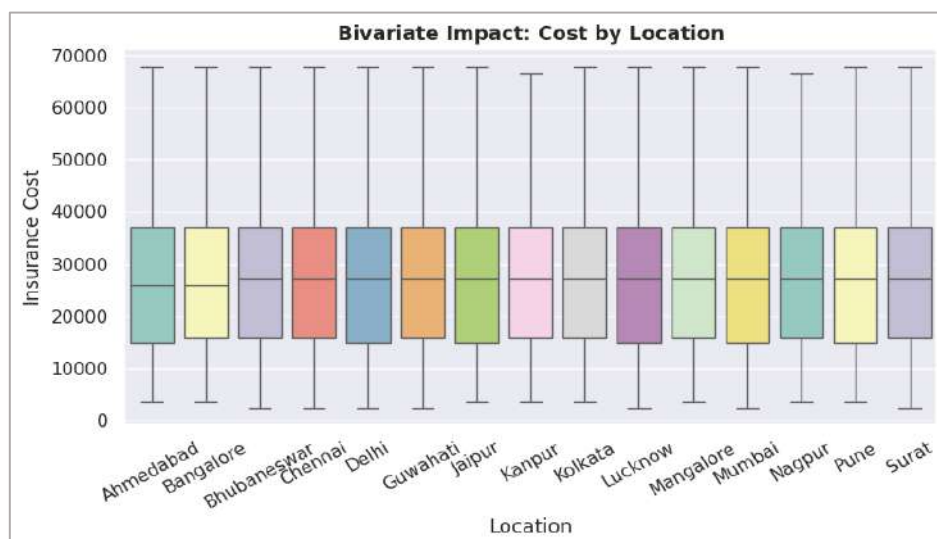


Figure 42. Appendix- Boxplot of Location vs insurance_cost

- All 15 cities have roughly comparable median insurance costs, ranging from 25,000 to 27,000. There is no notable relationship between location and insurance costs, suggests this column provides no predictive value to the model because there is no regional pricing signal. We can safely drop this column in data pre-processing sections.

Bivariate Analysis on Categorical Feature vs Features (Clinical Inter- Relationships)

We used a scatter plot here to visualize the feature vs feature distribution and variability of clinical outcomes across different categories, in order to analyse clusters and outliers that summary statistics might have hidden.

Fat Percentage vs Average Daily Steps

This plot asses how different levels of physical activity influence fat percentages, reflecting lifestyle implications for individual's health.

- Fat percentage ranges from around 11 to 42, while average daily steps vary between 2000 and 11000, indicates data points are widely scattered with no visible upward or downward trend. This implies that there is no clear relationship with daily physical activity levels and fat percentage, which contrasts with the expectation that higher fat percentage might be associated with lower physical activity.

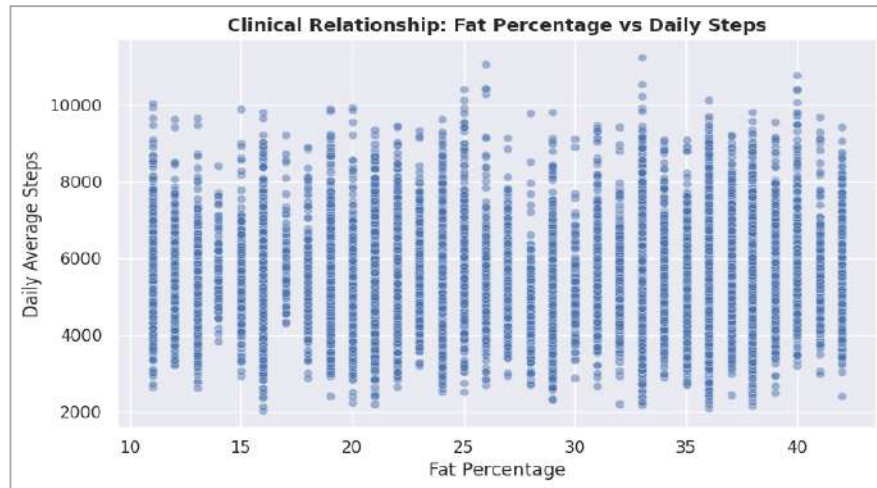


Figure 43. Appendix -Scatter plot of fat_percentage vs daily_avg_steps

Fat Percentage vs Average Glucose Level

It analyses the implications of body fat percentages on glucose levels to assess potential individual related health risks.

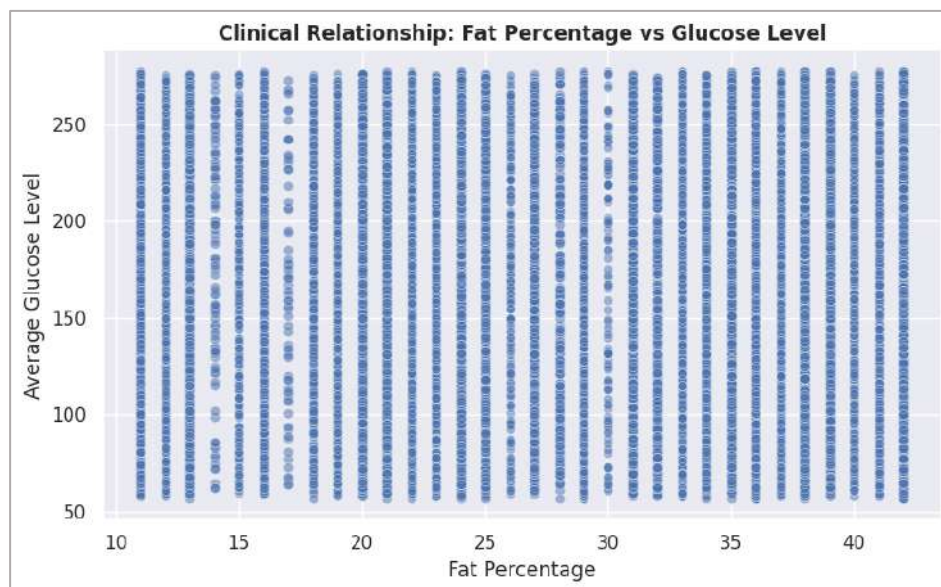


Figure 44. Appendix -Scatter plot of fat_percentage vs avg_glucose_level

- Fat percentage ranges approximately from 11 to 42, while average glucose levels span from around 60 to 270, with no clear linear trend and no distinct clustering pattern. This means that there isn't a strong or direct link between average glucose levels and fat percentage.

BMI vs Gender

To analyses the implications of gender on bmi to assess how individual's bmi vary based on gender.

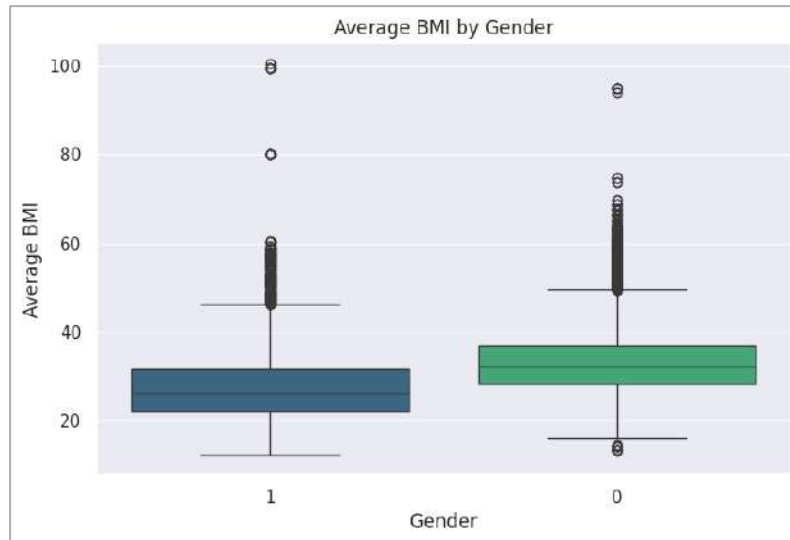


Figure 45. Appendix-Scatter plot of bmi vs Gender

- Medians (Female 1 vs. Male 0): The female median BMI is approximately 26. At 32, the median BMI for men is marginally higher. Several outliers beyond the whiskers are seen in both sexes. The highest outliers have BMIs of 95 for men and 100 for women. Higher fat percentages is loosely associated to lower step counts, which is consistent with expectations but offers a weak signal.

Bivariate Analysis on Categorical vs Categorical Features

Here, I will use a stacked bar plot along to show how different categories break down into sub-category, making it easy to see both the total counts and the relative mix of outcomes within each group using cross- tab.

Gender vs. Smoking Status

It analyses different levels of smoking risks based on gender to assess individual's health risk.



Figure 46. Stacked bar plot of Gender vs smoking_status

- Exercise distribution is fairly uniform among Salaried, Business, and Student groups; profession does not influence exercise habits.

11.4 Extended Multivariate Analysis

We performed multi-variate analysis to understanding how multiple factors together influence insurance cost. Extended multivariate analysis includes interaction effects that were not included in EDA Section 3.4.

Interaction Effect Plots (Multi-variable Relationships):

To understand how the relationship between the predictor and the target variable changes when the third variable is introduced, Interaction Effect Plots are used.

These plots are used for:

- Understanding the overall effect of lifestyle and clinical factors together
- Determining if certain groups, such as smokers and non-smokers, behave differently under the same conditions
- Uncovering patterns that are not visible through bivariate analysis

BMI vs Insurance Cost vs Exercise

This plot is used to assess whether people with similar BMI are charged differently based on their level of exercise.

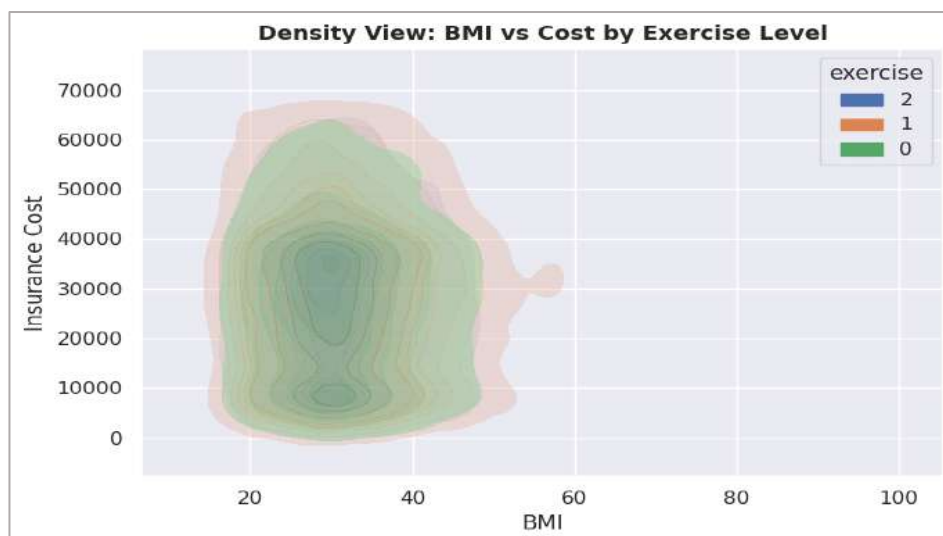


Figure 47. Density plot of bmi vs insurance_cost vs exercise

- Exercise Level 2 leaves the smallest mark on the graph. People who work out often stay in the standard cost range (under 40,000). Patterns from BMI and exercise frequency reveal BMI as the main driver for costs. People with BMI over 45 don't manage to cut their insurance expenses by ramping up exercise. Exercise levels don't do enough to separate cost ranges. Their central overlap blocks clear cost distinction unless we consider other features.

11.5 Extended Data Pre-Processing

Full statistical summary, unique value counts, and data quality check results are mentioned in Sections 2 and 4 of the main report.

Data Cleaning

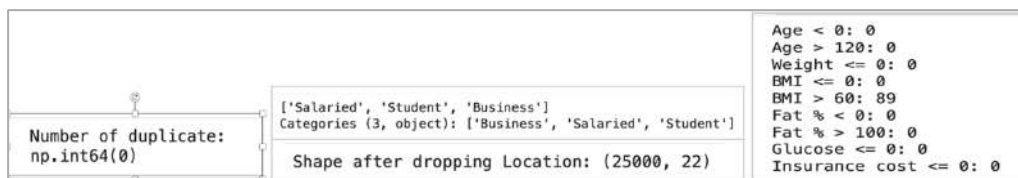


Figure 48. Appendix-Data cleaning outputs

Missing Values Treatment

```
Missing values remaining:
Series([], dtype: int64)

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25000 entries, 0 to 24999
Data columns (total 23 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   years_of_insurance_with_us               25000 non-null  int64
1   regular_checkup_last_year                25000 non-null  int64
2   adventure_sports                         25000 non-null  int64
3   Occupation                               25000 non-null  category
4   visited_doctor_last_1_year              25000 non-null  int64
5   cholesterol_level                       25000 non-null  category
6   daily_avg_steps                          25000 non-null  int64
7   age                                       25000 non-null  int64
8   heart_decs_history                      25000 non-null  int64
9   other_major_decs_history                25000 non-null  int64
10  Gender                                   25000 non-null  category
11  avg_glucose_level                       25000 non-null  int64
12  bmi                                       25000 non-null  float64
13  smoking_status                           25000 non-null  category
14  weight                                   25000 non-null  int64
15  covered_by_any_other_company             25000 non-null  category
16  Alcohol                                  25000 non-null  category
17  exercise                                  25000 non-null  category
18  weight_change_in_last_one_year           25000 non-null  int64
19  fat_percentage                           25000 non-null  int64
20  insurance_cost                           25000 non-null  int64
21  ever_hospitalized                        25000 non-null  int64
22  years_since_last_admission               25000 non-null  int64
dtypes: category(7), float64(1), int64(15)
memory usage: 3.2 MB
None
```

Table 29. Appendix- Missing Values

BMI Outlier Treatment

BMI before treatment: Min=12.30, Max=100.60
BMI after treatment : Min=12.55, Max=48.95
BMI outliers remaining: 0

Table 30. Appendix- Outliers Treatment

Feature Engineering

New feature “ever_hospitalized”:

- Balanced distribution:** 52% hospitalised (13,119); 48% never (11,881), indicating that this is not random missingness but a structurally relevant division. The significant difference in tenure distribution between groups (0-1 years vs 3+ years) strongly shows that a hospitalisation history becomes more frequent with longer policy tenure. This pattern may reflect either increased

likelihood of hospitalization over longer tenure or better documentation of medical history for long-term customers.

```

ever_hospitalized distribution:
ever_hospitalized
1    13119
0     11881
Name: count, dtype: int64

ever_hospitalized distribution (normalized):
ever_hospitalized
1     0.52
0     0.48
Name: proportion, dtype: float64

ever_hospitalized vs years_of_insurance_with_us:
years_of_insurance_with_us
ever_hospitalized
0      0.25  0.24  0.15  0.06  0.06  0.06  0.06  0.06
1      0.00  0.00  0.00  0.17  0.17  0.17  0.16  0.17

years_of_insurance_with_us
ever_hospitalized
0      0.06
1      0.17

ever_hospitalized vs covered_by_any_other_company:
covered_by_any_other_company
ever_hospitalized
0      0.8  0.2
1      0.6  0.4

```

Table 31. Appendix- ever_hospitalized distributions across features

Creating New Feature “health_risk_score”:

```

health_risk_score value counts:
health_risk_score
0     10169
1     12890
2      1774
3       167
Name: count, dtype: int64

```

Table 32. Appendix-New Feature health_risk_score value counts

- The health_risk_score runs from 0 to 3, with 0 representing no medical risk factors and 3 suggesting the presence of all three risk factors simultaneously. The distribution indicates a clear downward trend, as the medical risk grows from 1 to 3, fewer people fall into the higher-risk category, which corresponds to real-world health demographics.

Class Imbalance Check

- The **target variable "insurance_cost" is continuous** in nature, with values ranging from 2,468 to 67,870 and has an average of around 27,147. Since the target variable is continuous & variable in nature (a regression problem), indicating **class imbalance handling is not applicable, and no treatment required in this case.**

```

Target variable (insurance_cost) type: int64
-----
count    25000.000000
mean     27147.407680
std      14323.691832
min       2468.000000
25%      16042.000000
50%      27148.000000
75%      37020.000000
max      67870.000000
Name: insurance_cost, dtype: float64

```

Table 33. Appendix-Class Imbalance Check

Data preparation for Modeling

Encoding:

```
Remaining object columns: []
Shape after encoding: (25000, 25)

-----
Total missing values: 0

-----
Columns after encoding:
['years_of_insurance_with_us', 'regular_checkup_lasy_year', 'adventure_sports', 'visited_doctor_last_1_year', 'cholesterol_level', 'daily_avg_steps',
'percentage', 'insurance_cost', 'ever_hospitalized', 'years_since_last_admission', 'health_risk_score', 'Occupation_Salaried', 'Occupation_Student']
```

Table 34. Appendix-Encoding Columns

Train-Test Split:

We split the data in following set:

- X_train → 80% of feature data used to train the model.
- X_test → 20% of feature data used to test (evaluate) the model.
- y_train → Target values for X_train.
- y_test → Target values for X_test.

```
X_train shape : (20000, 24)
X_test shape  : (5000, 24)
y_train shape : (20000,)
y_test shape  : (5000,)
```

Table 35. Appendix- Train-Test Split

Pre-scaling sanity check:

```
=====
PRE-SCALING CAUTION CHECK
=====
Columns being scaled:
['years_of_insurance_with_us', 'regular_checkup_lasy_year', 'adventure_sports', 'visited_doctor_last_1_year', 'cholesterol_level', 'daily_avg_st
Total numeric columns: 24
All numeric columns exist in X_test

Data Quality Check:
years_of_insurance_with_us: Clean
regular_checkup_lasy_year: Clean
adventure_sports: Clean
visited_doctor_last_1_year: Clean
cholesterol_level: Clean
daily_avg_steps: Clean
age: Clean
heart_decs_history: Clean
other_major_decs_history: Clean
Gender: Clean
avg_glucose_level: Clean
bmi: Clean
smoking_status: Clean
weight: Clean
covered_by_any_other_company: Clean
Alcohol: Clean
exercise: Clean
weight_change_in_last_one_year: Clean
fat_percentage: Clean
ever_hospitalized: Clean
years_since_last_admission: Clean
health_risk_score: Clean
Occupation_Salaried: Clean
Occupation_Student: Clean

=====
Ready to scale!
=====
```

Table 36. Pre-scaling sanity check

Feature Scaling:

```
X_train mean after scaling (should be ~0): 0.0
X_train std after scaling (should be ~1): 1.0
X_test mean after scaling (should be ~0): -0.0052
X_test std after scaling (should be ~1): 1.0005
```

Table 37. Appendix-Feature Scaling

Final Data Verification & Data leakage Check Before Modeling:

This is a final verification step, it ensures dataset is completely clean, fully encoded, correctly shaped, and ready for baseline model training.

Final dataset shape	: (25000, 25)
X_train shape	: (20000, 24)
X_test shape	: (5000, 24)
Any nulls in X_train	: 0
Any nulls in X_test	: 0
Any nulls in y_train	: 0
Any nulls in y_test	: 0
Feature columns	: ['years_of_insurance_with_us', 'regular_checkup_lasy_year', 'adventure_sports', 'visited_doctor_last_1_year', 'cholesterol']
Skewness check before log transformation decision:	
insurance_cost	0.331650
avg_glucose_level	-0.006389
bmi	0.446274
daily_avg_steps	0.908867
dtype:	float64

Table 38. Appendix-Final Data Verification Before Modeling

We perform this step to record all leakage prevention techniques implemented during data pre-processing to guarantee that model evaluation accurately represents genuine out-of-sample generalisation performance.

=====
DATA LEAKAGE PREVENTION SUMMARY
=====
1. StandardScaler fit only on X_train:
-> Test statistics never influence training
2. No features derived from insurance_cost:
-> Target leakage fully prevented:
3. ever_hospitalized created from missingness pattern:
-> Not derived from target variable
4. Train-test split performed before scaling:
-> No cross-contamination between sets
5. smoking_status Unknown treated via median imputation:
-> No fake ordinal value introduced
6. No preprocessing step used full dataset statistics before split:
-> Full pipeline integrity maintained
X_test mean after scaling (should be ~0): -0.0052
X_test std after scaling (should be ~1): 1.0005
=====

Table 39. Appendix Data Leakage Handling

11.6 Full Model OLS Summary

Baseline Model-Linear Regression OLS Summary

OLS Regression Results						
Dep. Variable:	insurance_cost	R-squared:	0.945			
Model:	OLS	Adj. R-squared:	0.945			
Method:	Least Squares	F-statistic:	1.503e+04			
Date:	Tue, 31 Mar 2026	Prob (F-statistic):	0.00			
Time:	14:10:20	Log-Likelihood:	-1.9072e+05			
No. Observations:	20000	AIC:	3.815e+05			
Df Residuals:	19976	BIC:	3.817e+05			
Df Model:	23					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	-7.82e+04	356.700	-219.230	0.000	-7.89e+04	-7.75e+04
years_of_insurance_with_us	-89.8054	11.485	-7.819	0.000	-112.317	-67.293
regular_checkup_lasy_year	-375.7115	21.414	-17.545	0.000	-417.684	-333.739
adventure_sports	227.8323	86.724	2.627	0.009	57.846	397.819
visited_doctor_last_1_year	-51.5637	21.230	-2.429	0.015	-93.176	-9.952
cholesterol_level	23.1176	22.486	1.028	0.304	-20.956	67.191
daily_avg_steps	-0.0159	0.023	-0.689	0.491	-0.061	0.029
age	4.1352	1.476	2.802	0.005	1.243	7.028
heart_decs_history	198.3288	86.731	2.287	0.022	28.328	368.330
other_major_decs_history	-61.7377	72.164	-0.856	0.392	-203.185	79.710
Gender	-23.1304	54.939	-0.421	0.674	-130.815	84.554
avg_glucose_level	0.1357	0.378	0.359	0.720	-0.606	0.877
bmi	-3.1120	3.597	-0.865	0.387	-10.162	3.938
smoking_status	-27.3464	32.354	-0.845	0.398	-90.763	36.070
weight	1465.4391	3.464	423.102	0.000	1458.650	1472.228
covered_by_any_other_company	1197.6920	54.000	22.180	0.000	1091.848	1303.536
Alcohol	-3.9789	38.055	-0.105	0.917	-78.570	70.613
exercise	-21.1483	37.229	-0.568	0.570	-94.120	51.823
weight_change_in_last_one_year	173.1259	15.233	11.365	0.000	143.268	202.984
fat_percentage	-1.9489	2.881	-0.676	0.499	-7.596	3.698
ever_hospitalized	-54.4907	78.712	-0.692	0.489	-208.774	99.792
years_since_last_admission	51.2604	5.576	9.193	0.000	40.331	62.190
health_risk_score	82.1004	39.910	2.057	0.040	3.874	160.327
Occupation_Salaried	87.7531	69.124	1.270	0.204	-47.735	223.241
Occupation_Student	45.1142	61.985	0.728	0.467	-76.381	166.609
Omnibus:	528.441	Durbin-Watson:	1.968			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	604.868			
Skew:	0.366	Prob(JB):	4.51e-132			
Kurtosis:	3.436	Cond. No.	7.28e+18			
Notes:						
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.						
[2] The smallest eigenvalue is 1.07e-26. This might indicate that there are strong multicollinearity problems or that the design matrix is singular.						

Table 40. Appendix- Baseline Model-Linear Regression

Refining Baseline Model-Linear Regression

VIF score after dropping VIF=inf values:

	Feature	VIF
9	weight	26.482380
5	age	8.346416
3	visited_doctor_last_1_year	8.082455
7	avg_glucose_level	7.702460
0	years_of_insurance_with_us	4.755561
15	health_risk_score	3.919009
14	years_since_last_admission	3.576826
12	exercise	3.439675
13	weight_change_in_last_one_year	3.221929
4	cholesterol_level	2.767736
17	Occupation_Student	2.707922
11	Alcohol	2.511954
6	Gender	1.589479
10	covered_by_any_other_company	1.559698
1	regular_checkup_lasy_year	1.541739
16	Occupation_Salaried	1.493744
8	smoking_status	1.465574
2	adventure_sports	1.097661

Table 41. Appendix- Dropped High VIF Values

Dealing with High p-value:

```
Dropping 'Alcohol' (p-value = 0.86928)
Dropping 'Gender' (p-value = 0.85267)
Dropping 'avg_glucose_level' (p-value = 0.71625)
Dropping 'exercise' (p-value = 0.60452)
Dropping 'Occupation_Student' (p-value = 0.45853)
Dropping 'cholesterol_level' (p-value = 0.52121)
Dropping 'smoking_status' (p-value = 0.41583)
Dropping 'Occupation_Salaried' (p-value = 0.17172)
Dropping 'health_risk_score' (p-value = 0.08819)

All remaining features are statistically significant (p ≤ 0.05).

Final selected features:
['years_of_insurance_with_us', 'regular_checkup_lasy_year', 'adventure_sports', 'visited_doctor_last_1_year', 'age', 'weight', 'covered_by_any_other

Removed features (in order):
['Alcohol', 'Gender', 'avg_glucose_level', 'exercise', 'Occupation_Student', 'cholesterol_level', 'smoking_status', 'Occupation_Salaried', 'health_r

\Refined Model Performance:
R-squared: 0.94534
Adjusted R-squared: 0.94532
```

Figure 49. Appendix-Dropped high p-value features

Refined Model OLS summary:

After dropping the high p-value, i.e., statistically insignificant features, we refitted the model again to obtain refined model.

OLS Regression Results						
Dep. Variable:	insurance_cost	R-squared:	0.945			
Model:	OLS	Adj. R-squared:	0.945			
Method:	Least Squares	F-statistic:	3.842e+04			
Date:	Tue, 31 Mar 2026	Prob (F-statistic):	0.00			
Time:	14:10:23	Log-Likelihood:	-1.9072e+05			
No. Observations:	20000	AIC:	3.815e+05			
Df Residuals:	19990	BIC:	3.815e+05			
Df Model:	9					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	-7.835e+04	263.748	-297.064	0.000	-7.89e+04	-7.78e+04
years_of_insurance_with_us	-89.0148	10.822	-8.225	0.000	-110.226	-67.803
regular_checkup_lasy_year	-376.9852	20.525	-18.367	0.000	-417.216	-336.755
adventure_sports	227.0184	86.688	2.619	0.009	57.103	396.934
visited_doctor_last_1_year	-49.9291	20.913	-2.387	0.017	-90.921	-8.937
age	4.1375	1.475	2.804	0.005	1.246	7.029
weight	1465.0273	3.203	457.438	0.000	1458.750	1471.305
covered_by_any_other_company	1200.1339	53.860	22.282	0.000	1094.564	1305.704
weight_change_in_last_one_year	173.4514	15.173	11.432	0.000	143.711	203.191
years_since_last_admission	52.4447	3.608	14.537	0.000	45.373	59.516
Omnibus:	531.451	Durbin-Watson:	1.968			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	608.560			
Skew:	0.367	Prob(JB):	7.13e-133			
Kurtosis:	3.437	Cond. No.	956.			
Notes:						
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.						

Table 42. Appendix-Refined Model -Linear Regression

11.7 Extended Linear Regression Assumptions Checks

Test for Multicollinearity

Multicollinearity occurs when predictor variables in a regression model are correlated. There are different ways of detecting (or testing) multicollinearity. One such way is by using the Variance Inflation Factor, or VIF.

- If VIF is between 1 and 5, then there is low multicollinearity.

- If VIF is between 5 and 10, we say there is moderate multicollinearity.
- If VIF is exceeding 10, it shows signs of high multicollinearity.

final VIF values for refined model:		
	Feature	VIF
5	weight	16.889288
4	age	8.254900
3	visited_doctor_last_1_year	7.983753
0	years_of_insurance_with_us	4.408905
7	weight_change_in_last_one_year	3.070130
8	years_since_last_admission	2.608119
6	covered_by_any_other_company	1.556804
1	regular_checkup_lasy_year	1.510419
2	adventure_sports	1.096862

Table 43. Appendix-Multicollinearity Assumption Check

- The assumption of no perfect multicollinearity for the refined model is satisfied as the design matrix is full-rank and the condition number is 956. Even though "weight" shows high multicollinearity (VIF = 16.89), this does not bias predictions or reduce R^2 , and is acceptable given the prediction focused business goals.
- Therefore, we keep all remaining 9 significant variables and proceed to test the remaining linear regression assumptions.

Test for Linearity of Variables & Independence of Residuals

Why Test:

- Linearity refers to a straight-line relationship between two variables. Predictor variables must have a linear relationship with the dependent variable.
- The independence of the error terms (or residuals) is crucial. If the residuals are not independent, the confidence intervals for coefficient estimates will be reduced, leading us to wrongly conclude that a parameter is statistically significant.

Procedure:

- These assumptions can be checked by creating a plot of fitted values vs residuals. If the residuals do not follow a pattern, the model is linear and independent.
- Otherwise, the model shows evidence of nonlinearity, and the residuals are not independent.
- If the linearity and independence assumptions are not satisfied then we can try to transform the variables and make the relationships linear.

Inference:

- The plot indicates that the refined model accurately depicts the linear relationship between variables and insurance costs.
- **We see no pattern in the plot above. Hence, the assumptions of linearity and independence are satisfied.**

Test for Normality of Residuals

Normality test is done:

- The error terms, or residuals, should follow a normal distribution. If the error components are not normally distributed, the confidence interval for coefficient estimates may become overly large or narrow. When a confidence interval becomes unstable, it makes it harder to estimate coefficients using least squares minimisation. Non-normality indicates that a few unexpected data points must be thoroughly investigated in order to create a better model.

Normality check steps:

- The shape of the residuals histogram can provide a first understanding of normality.
- It can also be verified using a Q-Q plot of residuals. If the residuals have a normal distribution, they will result in a straight line plot, otherwise not.
- The Shapiro-Wilk test is another method for determining normality.
 - Null hypothesis: The residuals are regularly distributed.
 - Alternative hypothesis: The residuals are not normally distributed.
- If assumption failed then we can apply transformations such as log, exponential, and arcsinh, etc., to our data.

Normality of Residuals Plot

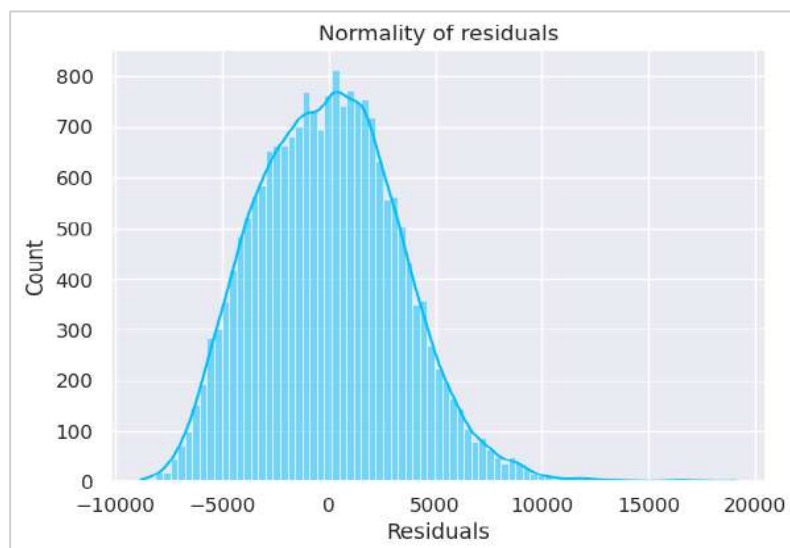


Figure 50. Appendix-Normality of Residuals Plot- Refined LR model

- The distribution of residuals is roughly symmetric and bell-shaped

Shapiro-Wilk test

The Shapiro-Wilk test can also be used for checking the normality. The null and alternate hypotheses of the test are as follows:

- Null hypothesis - Data is normally distributed.
- Alternate hypothesis - Data is not normally distributed.


```
Shapiro-Wilk test for Normality of residuals:
ShapiroResult(statistic=np.float64(0.9895707193758183), pvalue=np.float64(2.3783139041869752e-35))
```

Table 44. Appendix-Shapiro-Wilk Test

- The test statistic is 0.9896 and p-value < 0.05, around 2.38×10^{-35} , so we reject the null hypothesis.

Test for Homoscedasticity

Homoscedasticity: If the variance of the residuals is symmetrically distributed throughout the regression line, the data is considered homoscedastic. This is ideal because it ensures that the standard errors of the coefficients are accurately estimated, making confidence intervals and p-values reliable.

Heteroscedasticity: If the variance of the residuals across the regression line is unequal, the data is considered heteroscedastic. Presence of heteroscedasticity violates the assumption of constant variance. While it does not bias the coefficient estimates or predictions, it does increase or decrease standard errors, leading to potentially incorrect inference (e.g., declaring a variable significant when it is not).

Need for the test:

- The presence of non-constant variance in the residuals causes heteroscedasticity.
- Non-constant variance occurs in the presence of outliers.

checking for homoscedasticity:

- The residual vs fitted values plot can be used to test for homoscedasticity. Heteroscedasticity can result in non-symmetrical residual shapes, such as arrows. The Goldfeld-Quandt test can also be utilised. If the p-value exceeds 0.05, we can conclude that the residuals are homoscedastic. Otherwise, they're heteroscedastic.
 - Null hypothesis: The residuals are homoscedastic (constant variance).
 - Alternative hypothesis: Residuals exhibit heteroscedasticity (non-constant variance)
- If this assumption is not followed then heteroscedasticity can be reduced by incorporating more essential features or performing transformations.

```
Goldfeld-Quandt test for homoscedasticity:
[('F statistic', np.float64(0.9944083892896372)),
 ('p-value', np.float64(0.6103447563423754))]
```

Table 45. Appendix-Gold-Quandt Test

- We obtained F-statistic = 0.9944, p-value = 0.6103 from the test results.
- The p-value is greater than 0.05, so we cannot reject the null hypothesis of homoscedasticity.

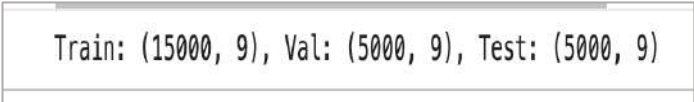
11.8 Final Baseline-Linear Regression Model-Extended

Final Dataset Preparation (Refined Features)- Train / Validation / Test split (60:20:20)

Methodology:

- The dataset is split into three mutually exclusive parts:
 - **Training set (60%)** - used to fit all models.
 - **Validation set (20%)** - used to compare model performance and tune hyperparameters.
 - **Test set (20%)** - kept completely unseen until the final model is selected.
- **Splitting the data into train, validation, and test sets (60:20:20 ratio):**
 - ***X_train_base* (60%)** - It is used to train the model.
 - ***X_val_base* (20%)** - It is used for model validation, hyperparameter tuning, and performance evaluation during development.
 - ***X_test_base* (20%)** - It is used for final, unbiased evaluation of the model after all tuning is complete.
 - ***y_train_base*** - Here the target values corresponding to *X_train_base*.
 - ***y_val_base*** - The target values for *X_val_base*.
 - ***y_test_base*** - Target values for *X_test_base*.
 - Here the data is first split into training (60%) and temporary (40%) sets. The temporary set is then equally divided into validation (20%) and test (20%) sets to maintain the required 60:20:20 ratio.

Why this step is necessary: A separate validation set prevents overfitting during model selection and tuning. The test set provides an unbiased estimate of the final model's real-world performance. This split respects the golden rule of machine learning: **test only once**.



Train: (15000, 9), Val: (5000, 9), Test: (5000, 9)

Figure 51. Appendix-Train / Validation / Test split (60:20:20)

Feature Scaling (Standardization) for Final Baseline Model

Methodology: A StandardScaler is fitted only on the training set and then used to transform the validation and test sets. The scaler centers features to mean = 0 and scales to unit variance.

Why this step is necessary: Sklearn's Linear regression uses coefficient based optimisation. Thus, scaling is required for final baseline model. However, Tree-based models are scale-invariant i.e scaling does not change their predictions or splits. Therefore, we applying the same scaling to all models to keep the workflow consistent as it does not harm performance.

Scaled Training Data:			
	years_of_insurance_with_us	regular_checkup_last_year	adventure_sports
0	-0.033764	-0.648650	-0.298077
1	0.349631	-0.648650	-0.298077
2	0.733025	0.182953	-0.298077
3	0.733025	1.846158	-0.298077
4	-1.567344	-0.648650	-0.298077
	visited_doctor_last_1_year	age	weight
0	1.697729	0.375186	-0.930063
1	-0.077891	-1.681528	-0.070465
2	-0.077891	-0.871307	0.036984
3	-0.077891	0.936108	0.144434
4	-0.965701	-1.182931	-2.004559
	covered_by_any_other_company	weight_change_in_last_one_year	
0	1.530683	-1.497451	
1	1.530683	-1.497451	
2	-0.653303	-0.902658	
3	1.530683	-1.497451	
4	-0.653303	2.071308	
	years_since_last_admission		
0	-0.830291		
1	-0.830291		
2	0.839463		
3	-0.830291		
4	-0.830291		

Table 46. Appendix -Scaling for Final Baseline Model

11.9 Extended-Advanced Model

Decision Tree Regressor

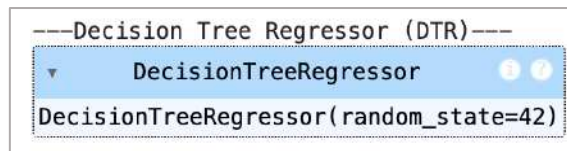


Table 47. Appendix- Decision Tree Regressor

- We have successfully build the decision tree model.

Random Forest Regressor

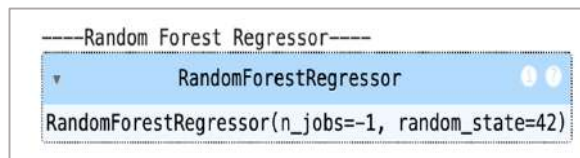


Table 48. Appendix-Random Forest Regressor

- We have successfully build the random forest model with default parameters.

Gradient Boosting Regressor

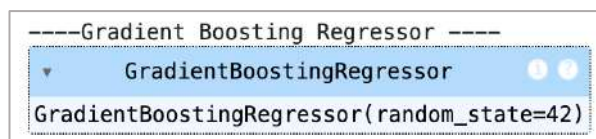


Table 49. Appendix- Gradient Boosting Regressor

- We have successfully build the gradient boosting regressor model with default parameters.

XGBoost Regressor

```
=== XGBoost Regressor ===
XGBRegressor
XGBRegressor(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=None, device=None, early_stopping_rounds=None,
              enable_categorical=False, eval_metric=None, feature_types=None,
              feature_weights=None, gamma=None, grow_policy=None,
              importance_type=None, interaction_constraints=None,
              learning_rate=0.1, max_bin=None, max_cat_threshold=None,
              max_cat_to_onehot=None, max_delta_step=None, max_depth=None,
              max_leaves=None, min_child_weight=None, missing=nan,
              monotone_constraints=None, multi_strategy=None, n_estimators=100,
              n_jobs=None, num_parallel_tree=None, ...)
```

Table 50. Appendix- XGBoost Regressor

- We have successfully build the XGBoost regressor model with default parameters.

AdaBoost Regressor

```
---- AdaBoost Regressor ----
AdaBoostRegressor
AdaBoostRegressor(n_estimators=100, random_state=42)
```

Table 51. Appendix- AdaBoost Regressor

- We have successfully build the AdaBoost regressor model with default parameters.

Performance comparison of all advance models

----- Advanced Models Comparison (Validation Set) -----					
Model	RMSE	MAE	R-squared	Adj. R-squared	MAPE (%)
XGBoost	2927.765701	2342.543701	0.958003	0.957927	11.499298
Gradient Boosting	2932.900850	2357.620479	0.957855	0.957779	11.720966
Random Forest	3083.159635	2455.617697	0.953426	0.953342	11.949105
AdaBoost	3256.976464	2696.569060	0.948027	0.947933	16.011246
Decision Tree	4212.695993	3258.253600	0.913050	0.912893	15.441546

Table 52. Appendix- Advance Models Performance Comparison

12 ADVANCED MODEL HYPERPARAMETER TUNING

The top 3 models (**XGBoost, Gradient Boosting, and Random Forest**) were chosen for further optimisation based on validation performance (RMSE being the primary metric).

Hyperparameter Tuning Strategy: Hyperparameter tuning is used to find the best parameter combinations that optimise predictive accuracy, generalisation, and model complexity. This stage verifies that the selected models performs best for the given dataset.

GridSearchCV was used for hyperparameter tuning to systematically evaluate all parameter combinations and ensure optimal model performance. This approach provides more reliable and reproducible results for the given data size. While RandomizedSearchCV is widely used in large-scale industrial settings for efficiency, GridSearchCV was chosen for this project because to its manageable parameter space and the need for a thorough and interpretable evaluation.